Appendix 3D: Results of Spatial Stock Assessment Models for Alaska Sablefish

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Introduction

Alaska sablefish (*Anoplopoma fimbria*) are currently managed as a single panmictic stock across the Gulf of Alaska (GOA), Bering Sea (BS), and Aleutian Islands (AI), informed by an age-and sex-structured statistical catch-at-age model. While the operational assessment assumes that the stock is spatially homogenous, research over the past several decades, along with knowledge of fishery dynamics, suggest that spatial structure likely exists to some degree. For instance, individuals have been shown to demonstrate age-specific (ontogenetic) movement patterns, wherein younger individuals reside in the BS and AI, and move towards the GOA as they mature (Cheng et al., 2025; Heifetz and Fujioka, 1991). In recent years, large recruitment events of age-2 individuals have also generally been found to occur in western regions (BS and AI). Moreover, fishery effort is primarily concentrated in the GOA regions, likely resulting in differential harvest impacts across space. Taken together, these patterns suggest that spatial heterogeneity in recruitment, movement, and harvest dynamics likely plays a role in shaping Alaska sablefish population dynamics.

One approach to improving understanding of spatial population dynamics is the use of spatially-explicit stock assessment models. In general, these models resemble single-region stock assessment models, but data are typically disaggregated across spatial regions, and movement dynamics are often incorporated into population processes. Importantly, spatial models often involve an increase in the number of parameters estimated, coupled with a decrease in the amount of data informing parameter estimates through regional partitioning of data. Thus, spatial stock assessment models are often subject to greater tensions between model complexity and data availability, raising the practical question of whether the gains in structural realism directly translate to improved management performance relative to simpler alternatives (i.e., single-region models).

Despite the potential increase in model complexity and instability relative to single-region alternatives, several motivations exist for developing spatially-explicit stock assessment models in the case of sablefish. First, spatial models can provide a more mechanistic understanding of regional population dynamics and allow monitoring of localized depletion, which are not made explicit from a panmictic stock assessment model. Second, spatial models can serve as conditioning frameworks for operating models used in management strategy evaluations (MSEs). Lastly, spatial models potentially provide a means to address spatial changes in survey design, allowing the accommodation of missing data, and circumventing the need to interpolate missing data, as would be the case in a single-region model. In the following, we present the development of two alternative spatial stock assessment models for Alaska sablefish.

Methods

General Model Structure

Spatial stock assessment models were developed using the SPoRC (Stochastic Population over Regional Components) modelling framework (Cheng et al., 2026). SPoRC is a generalized statistical catch-at-age model written in RTMB (Kristensen et al., 2016) with the ability to support multi-region, multi-fleet, multi-sex configurations, while also integrating tagging data. Two configurations of spatial models were developed: a three-region model and a five-region model. The three-region model partitions the stock across the BS, AI, and GOA, while the five-region model further disaggregates the GOA into the western GOA (WGOA), the central GOA (CGOA), and the eastern GOA (EGOA). Both configurations share common population dynamics and model specifications, differing primarily in spatial resolution and the number of estimated parameters. The spatial assessment models were based partly on the dynamics of the single region (or fleets as areas) model, the conceptual model shown in Figure 3D.1, and the parameterization of pervious work (Cheng et al., 2025). In particular, spatial models represented 30 ages (2 – 31+), sex-specific dynamics (females and males), 65 years (1960 – 2025), two fishery fleets (longline and trawl), and two survey fleets (domestic and Japanese longline surveys). Within each annual time step, the population was subject to the following processes in sequence: recruitment of age-2 individuals, movement among regions, followed by mortality from fishing and natural mortality. Further details on model dynamics, observation equations, and likelihoods are described in https://chengmatt.github.io/SPoRC/articles/c_model_equations.html.

Recruitment

Recruitment in both spatial models were assumed to follow a mean recruitment relationship, where global mean recruitment is apportioned across spatial regions. Lognormal recruitment deviations are estimated to vary across space and time, with the standard deviation of recruitment deviations fixed at 1.2 across all regions. Recruits were assumed to be unable to move, in order to mitigate potential confounding with movement and recruitment dynamics.

Natural Mortality

Natural mortality is sex-specific and assumed to be constant across sexes and ages, consistent with the single-region and fleets as areas models presented. Both values are fixed according to those estimated in the fleets as areas model (*26.2b_FAA_3Flt_LLS_Only*).

Growth and Maturity

Weight-at-age and maturity-at-age are treated as fixed quantities and are assumed to be constant across space, where values are taken from the operational assessment. Weight-at-age is assumed to be sex-specific and follows a von Bertalanffy growth function (females asymptote at a larger size than males), while maturity-at-age is only modelled for females and is represented as a logistic function.

Movement

Movement among regions was modelled as a discrete time Markovian process, assuming constant rates across time, but age-specific movement. Age-specific movement was represented as three age-blocks represented young individuals (ages 2 – 7), intermediate aged individuals (8 – 15), and old individuals (16 – 31) to approximately capture ontogenetic shifts in spatial distribution (i.e., movement from western to eastern regions as individuals matured).

Selectivity and Catchability

Across all fishery and survey fleets, selectivity was parameterized as sex-specific logistic functions. The longline fleet estimated time-varying selectivity as two time-blocks to capture a change in fleet structure (i.e., shifting from hook-and-line gear to pot gear), while the trawl fleet estimated time-invariant selectivity. Both survey fleets similarly estimated time-invariant selectivity. Selectivity was assumed to be constant across regions. Catchability for both survey fleets were freely estimated and assumed to remain constant across space and time.

Tag Reporting

Tag reporting rates were to characterize tag recaptures. Tag reporting was assumed to be constant across regions, but was allowed to vary across time as time blocks. In particular, three time-blocks were estimated, coinciding with large changes in fishery dynamics (inception of an individual fishing quota fishery and a change in fleet structure).

Observation Models and Likelihoods

Catch

Region-specific expected removals (weight) by the longline and trawl fishery are predicted using Baranov's catch equation, and are fit to assuming a lognormal likelihood with negligible error (standard deviation of catch is assumed to be fixed at 0.05).

Survey Indices

Region-specific survey indices (numbers) are fit to for both the domestic and Japanese longline survey fleets assuming a lognormal likelihood and design-based estimates of uncertainty. Unlike the single-region models, regional indices are fit directly without the need for interpolation or carryover values, and is able to adequately accommodate missing data.

Compositions

Both age and length-compositions are fit to assuming a Multinomial likelihood, where input sample sizes are generally based on *a priori* knowledge of sampling practices (i.e., regions and fleets with higher sampling coverage have larger sample sizes). Sex-specific compositions are available for all fishery and survey fleets and are fit to jointly (i.e., considering sex-specific compositions as a single vector; Cheng et al., 2025). Unlike the single-region counterparts, no data-weighting is performed to further adjust sample sizes, given complexities with respect to data-weighting in spatial models.

Tagging Data

Since the inception of the longline survey, approximately 400,000 individuals have been tagged, with about 35,000 individuals recaptured. Thus, tagging data are expected to be highly informative of movement dynamics. Expected tag recaptures are predicted using a Brownie tag attrition submodel, and are fit using a multinomial release-conditioned likelihood, where the probability of recapture in each region is derived from the model's predicted movement and survival processes. Moreover, the maximum tag liberty was restricted to 15 years (i.e., tags were not tracked or fit to beyond 15 years at liberty) to reduce computation time. To prevent tagging data from having due influence on overall model estimation, multinomial weights are scaled by a factor of 0.5.

Model Estimation and Diagnostics

All parameters were estimated using penalized maximum likelihood using RTMB, where the objective function was the sum of negative log-likelihoods for catch data, survey indices, compositions, tagging, penalties (i.e., recruitment), and priors placed on parameters. The three-region model estimates a total of 723 parameters, while the five-region model estimates a total of 1208 parameters. For both models, the largest share of parameters primarily results from fishing mortality deviations (~50%), followed by recruitment deviations (~27%), initial age deviations (~12%), and movement parameters (~5%).

Several diagnostic tools were utilized to assess model adequacy, stability, and performance. In particular, all models were inspected to ensure that maximum gradient components for all parameters were < 1e-5, as well as a positive definite Hessian matrix. Model fits to catch data, survey indices, both age- and length-compositions, and tagging data were also evaluated. Lastly, retrospective analyses (7-year peels) were conducted to understand model stability under sequential data removal and to evaluate retrospective consistency in recruitment and spawning stock biomass (*SSB*).

Results

Both spatial models converged with maximum gradient components for all parameters < 1e-5, along with an invertible Hessian matrix. In general, most data sources were adequately fit and did not demonstrate systematic misfits. For instance, both spatial models provided good fits to regional catch data (Fig. 3D.2 and 3D.3). Likewise, abundance indices from both survey fleets were generally fit well, although indices in the most recent years for the GOA regions appeared to be slightly overpredicted (Fig. 3D.4 and 3D.5). With respect to composition data, both models appeared to capture general patterns, wherein western regions often had higher proportions of younger fish, and eastern regions demonstrated higher proportions of older fish (Fig 3D.6 – 3D.43). While fits to composition data were generally adequate, there were some patterns in composition data that neither spatial model was able to reconcile. In particular, exceptionally large plus groups as observed by the longline fishery in the Aleutian Islands were underpredicted in some years (i.e., positive residuals for the plus group; Fig. 3D.7 and 3D.22). Relative to other data sources, tagging data were moderately well fit, although recaptures in recent periods appeared more difficult to reconcile (Fig. 3D.44 and 3D.45).

Retrospective analyses for both three-region and five-region models generally demonstrated similar patterns, where all regions exhibited positive retrospective patterns in *SSB* that often coincided with recruitment estimates from several years prior being overestimated (Fig. 3D.46 and 3D.47). Interestingly, recruitment estimates in the AI from the three-region model demonstrated some instability in certain retrospective peels (i.e., recruits in the Al are estimated at values close to 0), likely reflecting confounding (and revision of estimates) between movement and recruitment as age data were sequentially removed from the analysis (Fig. 3D.46). These results further highlight that spatially-explicit stock assessment models for Alaska sablefish will likely require careful annual tuning and evaluation of model structure, parameterization, and data weighting to ensure stable and biologically realistic dynamics through time.

Model estimates of recruitment and *SSB* were largely concordant between the spatial models (Fig. 3D.48). Both the pattern and scale of both recruitment and *SSB* were similar, with large recruitment events primarily occurring in western regions (BS and AI) and sustained recruitment events in the eastern regions (WGOA, CGOA, and EGOA) (Fig. 3D.48 and 3D.49). Moreover, regions exhibiting high recruitment did not necessarily coincide with regions of high SSB, presumably due to the estimated ontogenetic movement dynamics (i.e., younger individuals primarily residing in western regions and moving eastward as they mature), as well as potential unmodelled larval dispersal processes (Cheng et al., 2025) (Fig. 3D.48 and 3D.49).

Overall, results from the spatial model highlight that spatial heterogeneity in population dynamics likely exists, driven in part by movement dynamics, differential spatial harvest, and underlying oceanographic processes. Despite the presence of such heterogeneity, aggregated time-series estimates from spatial models generally align with simpler alternatives in the case of Alaska sablefish (e.g., the single-region model; Cheng et al., 2025), suggesting that these simpler alternatives provide similar advice management purposes. Importantly, retrospective analyses underscored the potential for instability in spatial models, and the need for further work to fine-tune these models. We propose that the developed spatial model serves as a useful research tool for: 1) understanding spatial population dynamics, and 2) serving as conditioning tools for simulation exercises.

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Figures

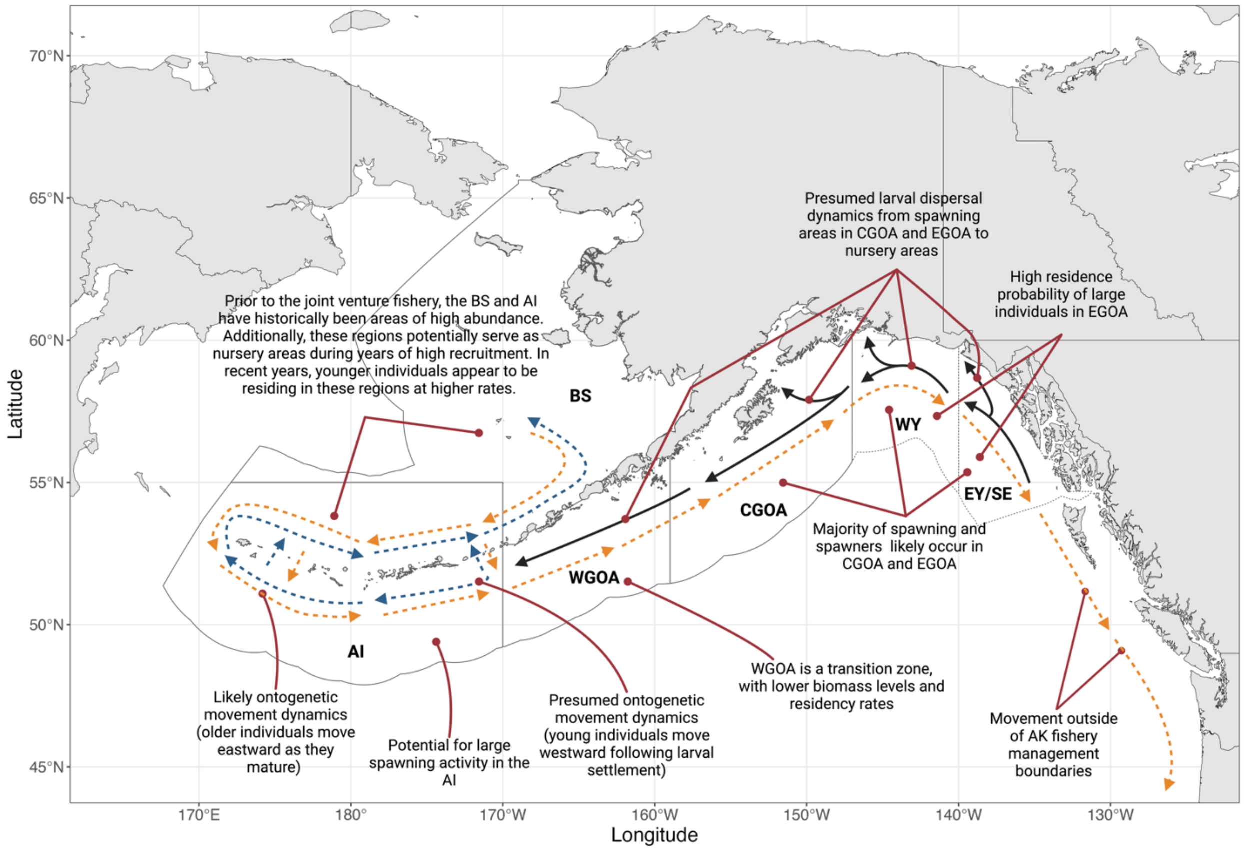

Figure 3D.1. Conceptual model of spatial sablefish dynamics. Black solid arrows represent hypotheses on larval dispersal, blue dashed arrows are hypotheses regarding the movement dynamics of younger individuals, and orange dashed arrows are hypotheses on general movement dynamics of individuals as they mature.

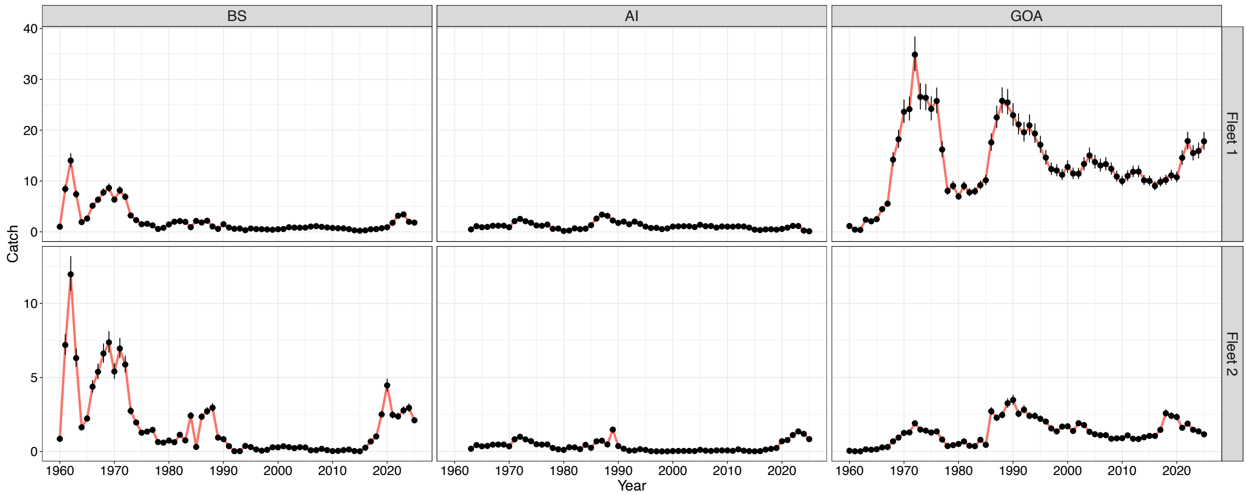

Figure 3D.2. Fits to regional catch data by fishery fleet for the three-region model. Fleet 1 represents the longline fishery while fleet 2 represents the trawl fishery. The redline denotes the model expectation and point ranges represents observed data and their associated 95% confidence intervals.

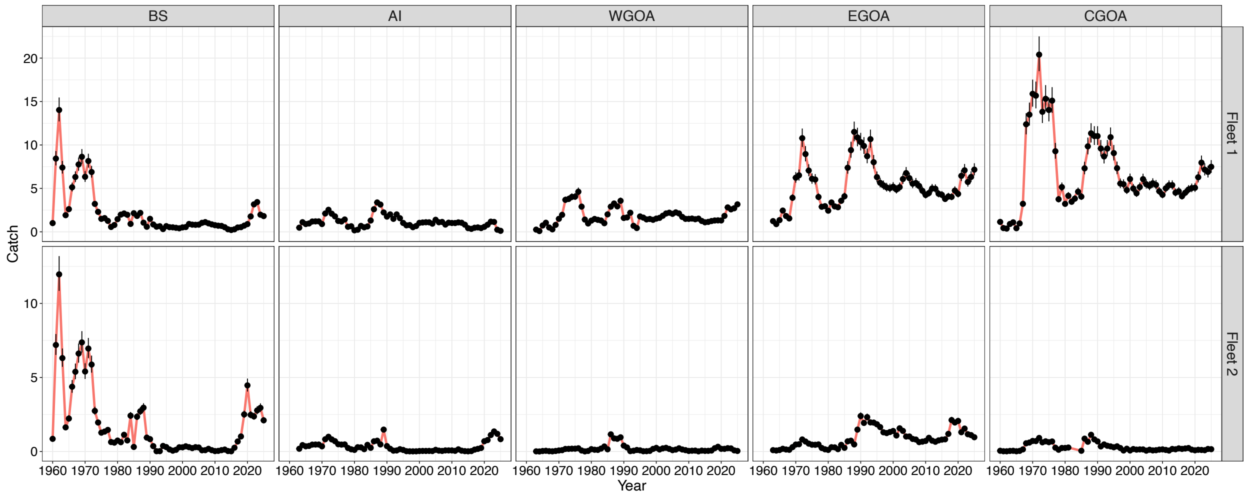

Figure 3D.3. Fits to regional catch data by fishery fleet for the five-region model. Fleet 1 represents the longline fishery while fleet 2 represents the trawl fishery. The redline denotes the model expectation and point ranges represents observed data and their associated 95% confidence intervals.

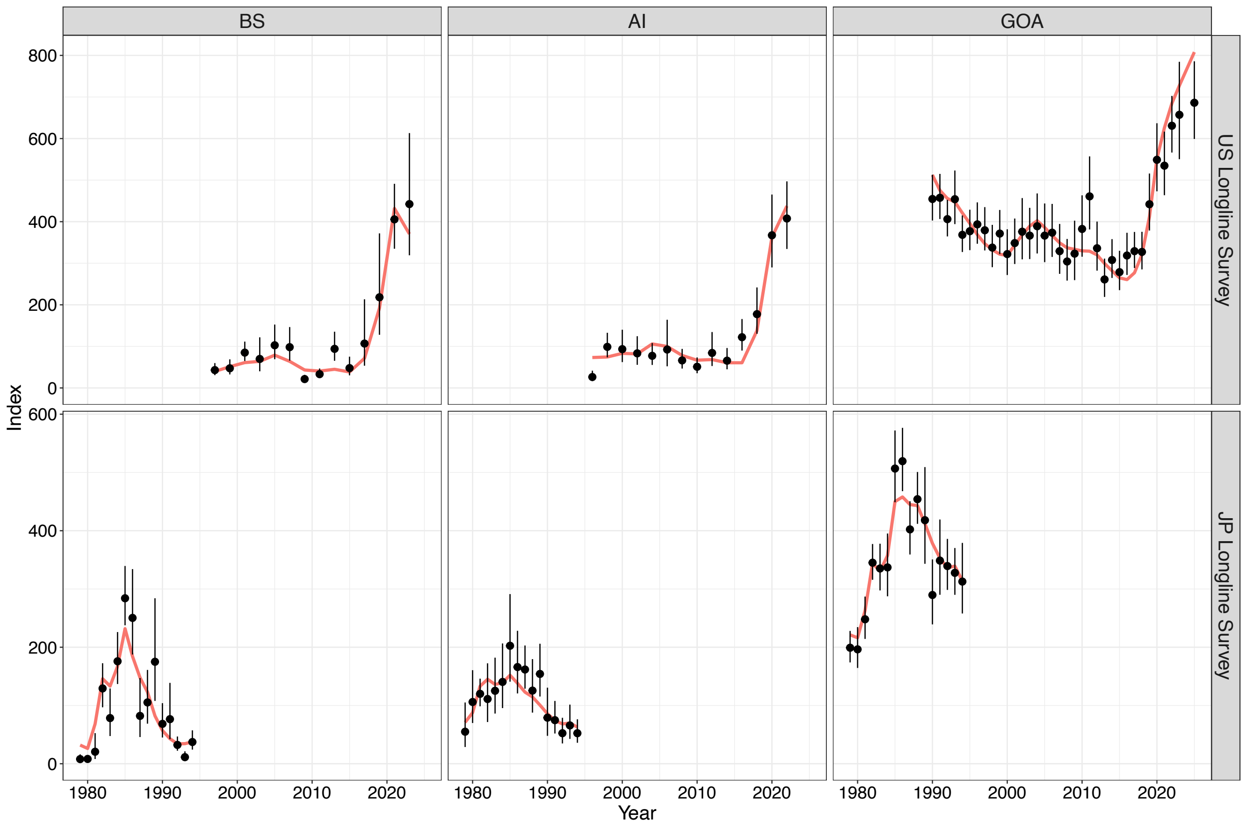

Figure 3D.4. Fits to regional survey indices by survey fleet for the three-region model. The redline denotes the model expectation and point ranges represents observed data and their associated 95% confidence intervals.

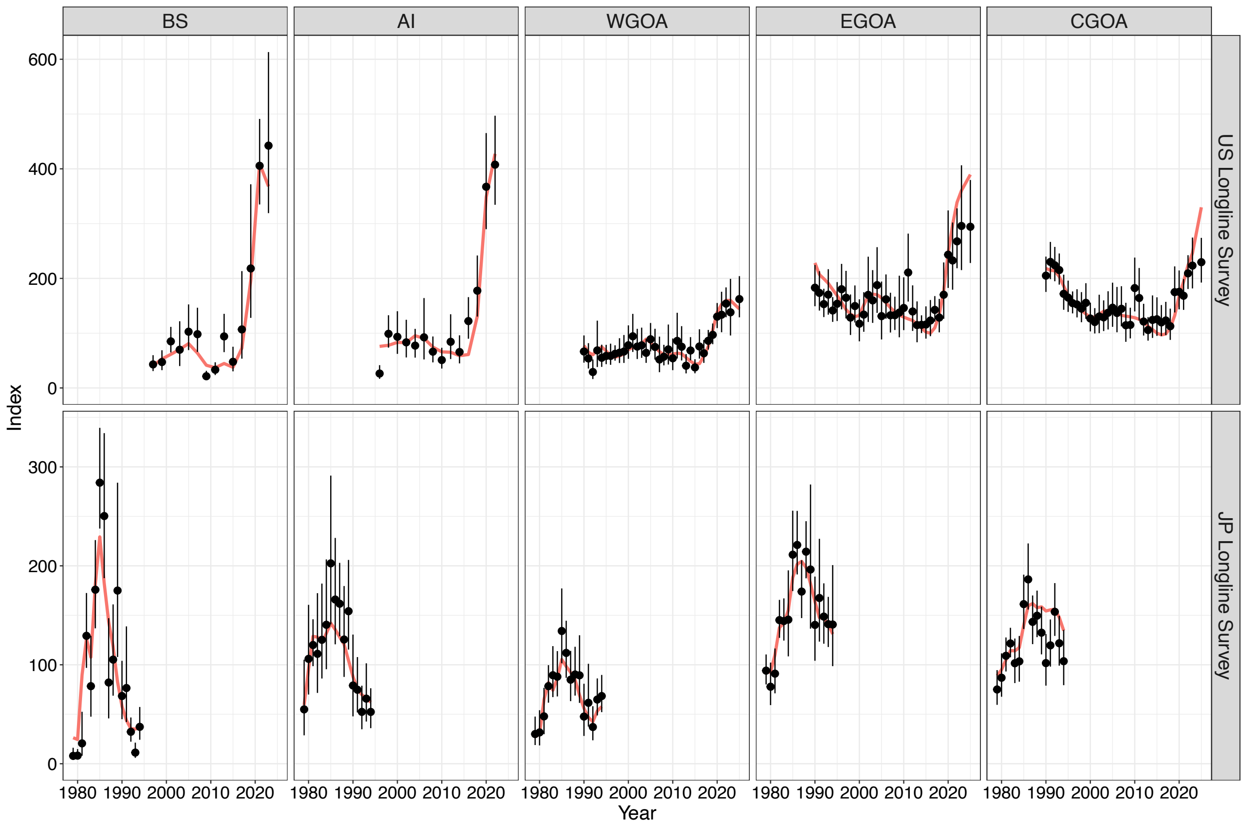

Figure 3D.5. Fits to regional survey indices by survey fleet for the three-region model. The redline denotes the model expectation and point ranges represents observed data and their associated 95% confidence intervals.

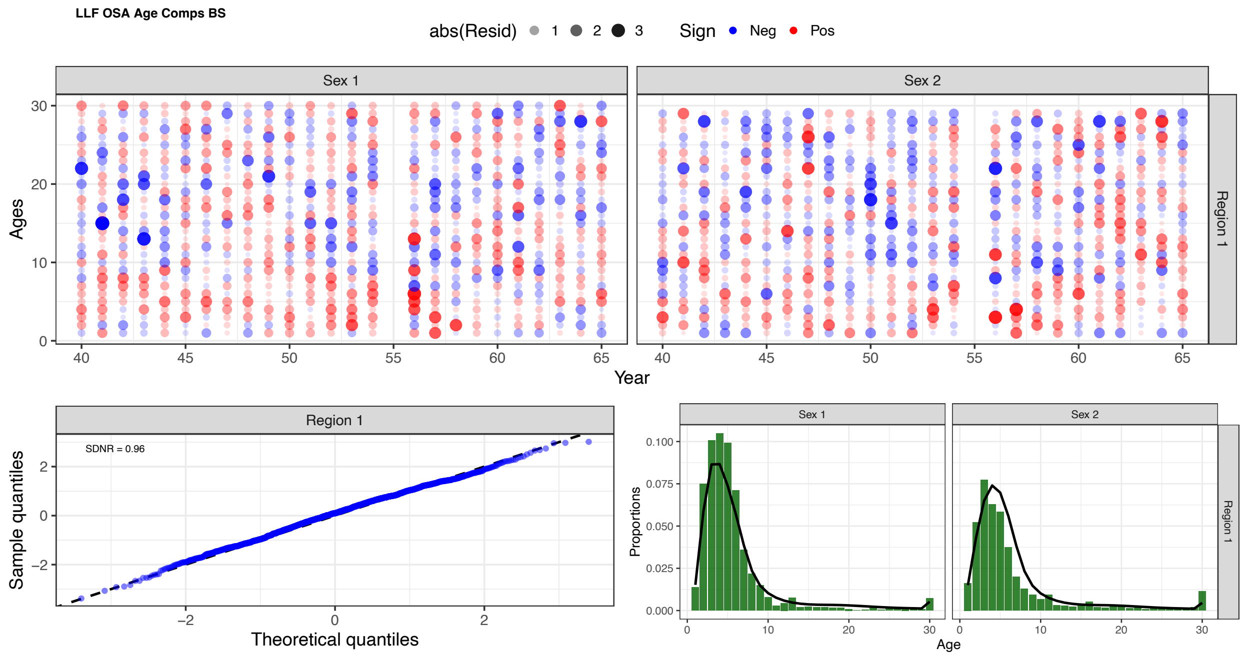

Figure 3D.6. Age composition fits from the three-region model for the longline fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

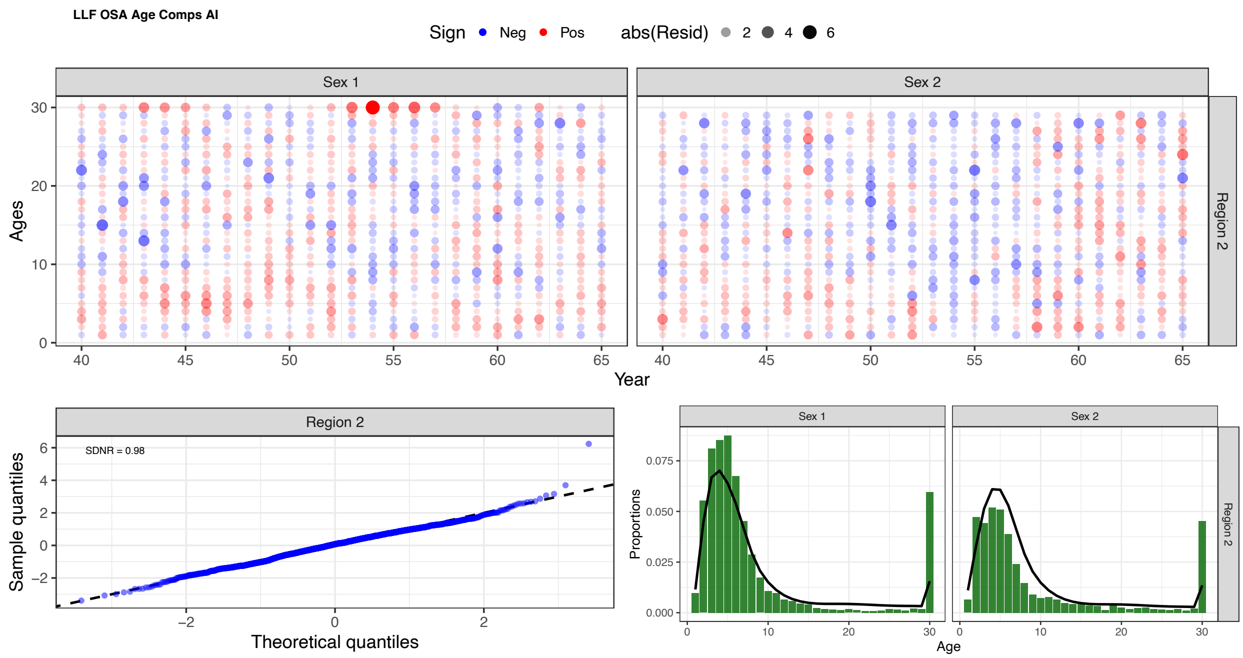

Figure 3D.7. Age composition fits from the three-region model for the longline fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

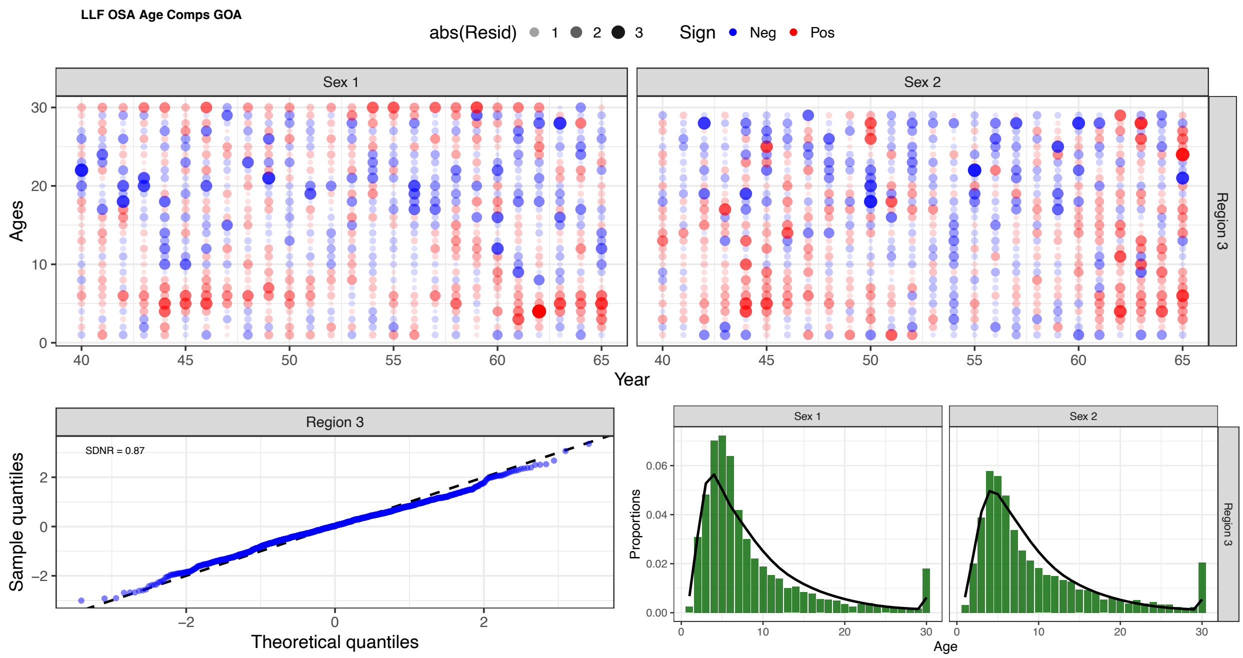

Figure 3D.8. Age composition fits from the three-region model for the longline fishery in the Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

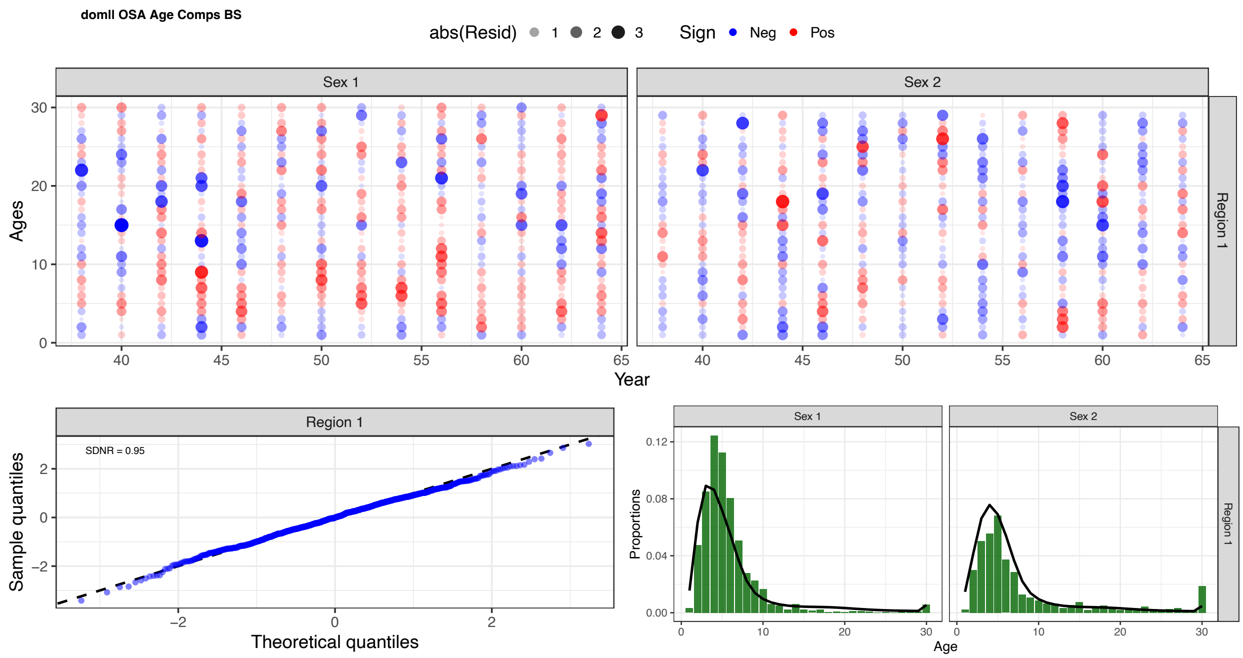

Figure 3D.9. Age composition fits from the three-region model for the domestic longline survey in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

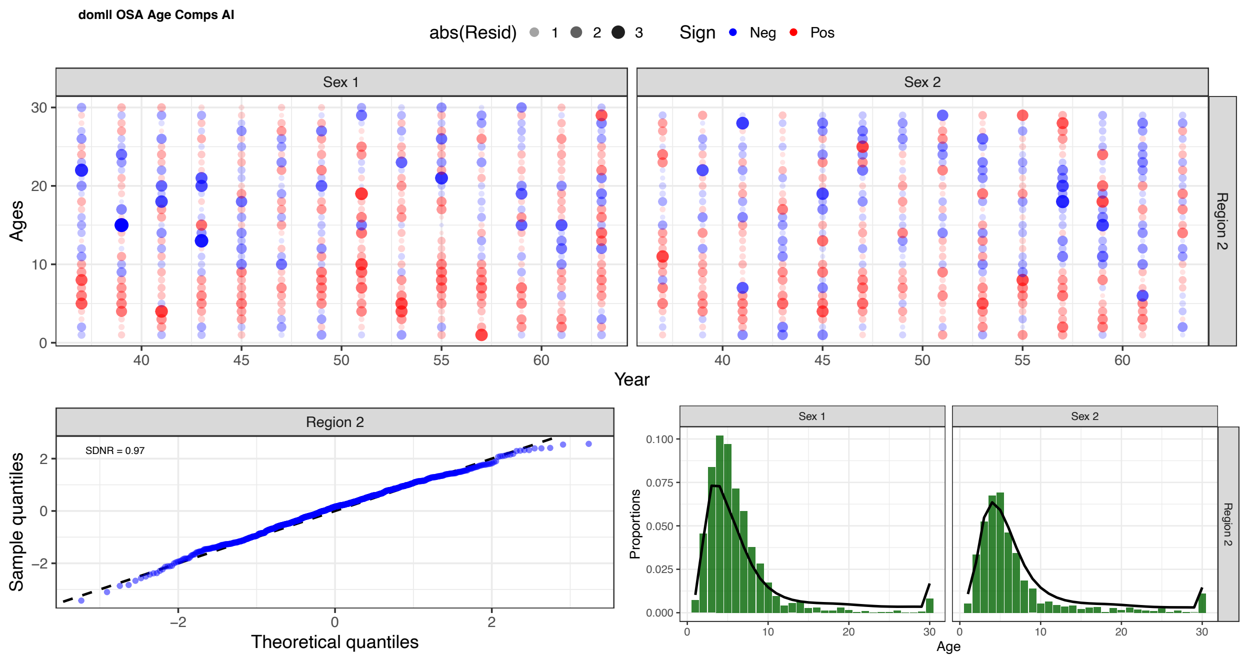

Figure 3D.10. Age composition fits from the three-region model for the domestic longline survey in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

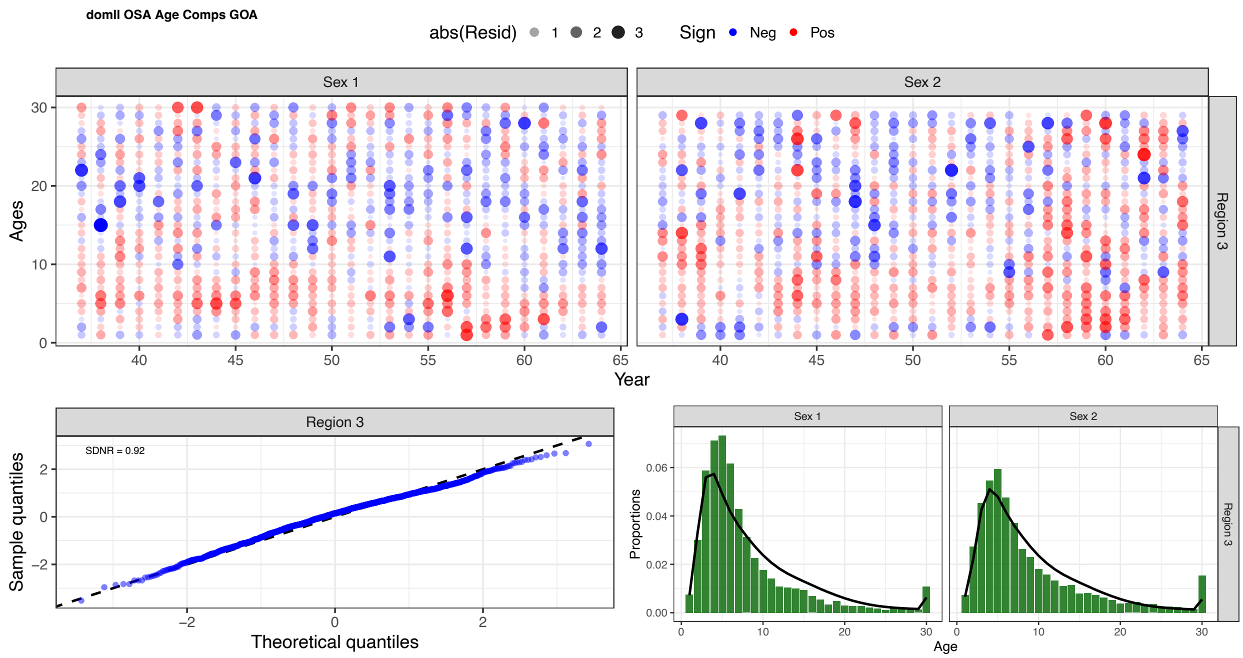

Figure 3D.11. Age composition fits from the three-region model for the domestic longline survey in the Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

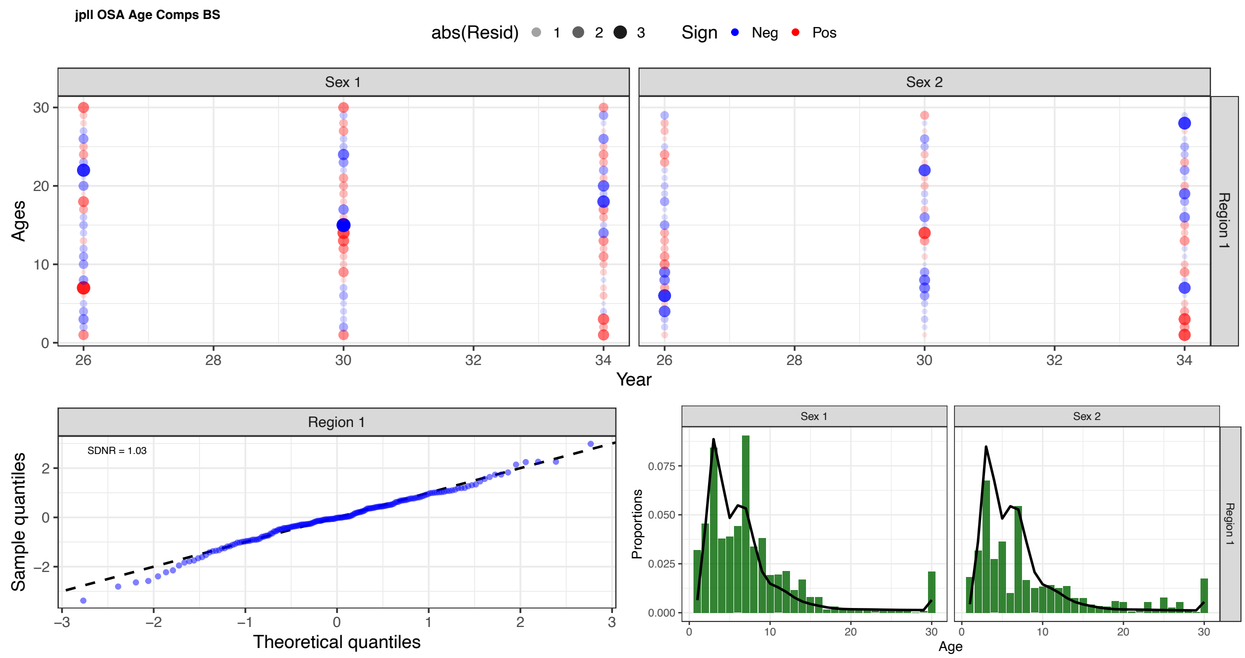

Figure 3D.12. Age composition fits from the three-region model for the Japanese longline survey in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

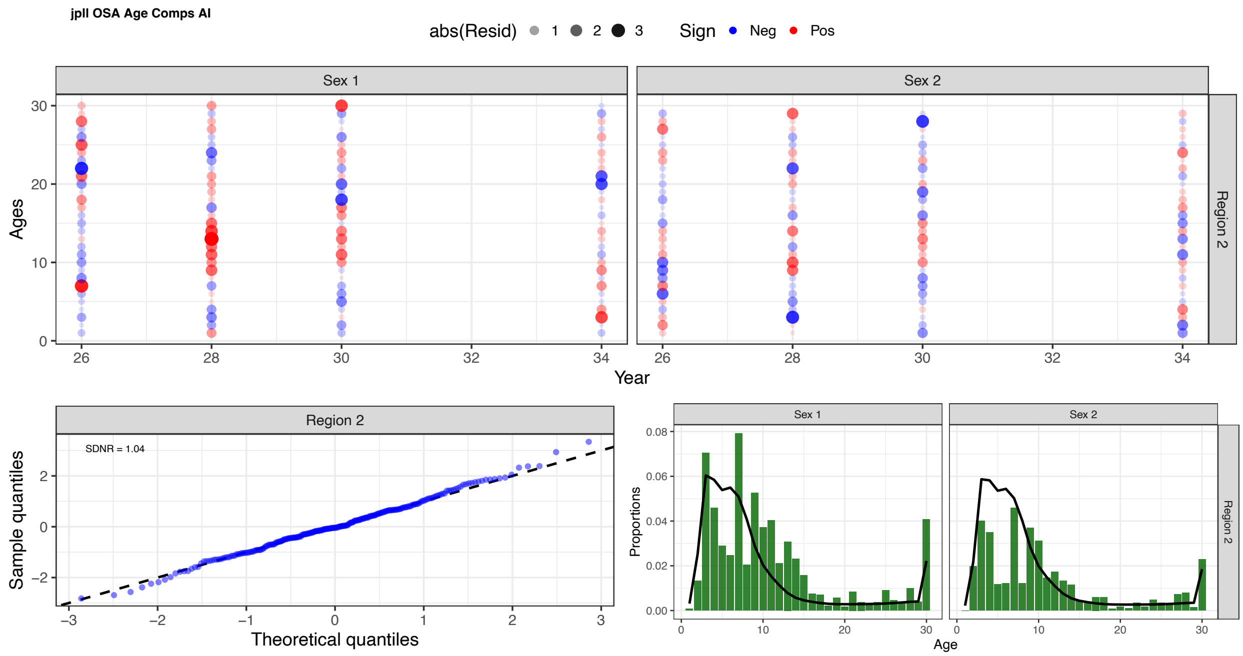

Figure 3D.13. Age composition fits from the three-region model for the Japanese longline survey in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

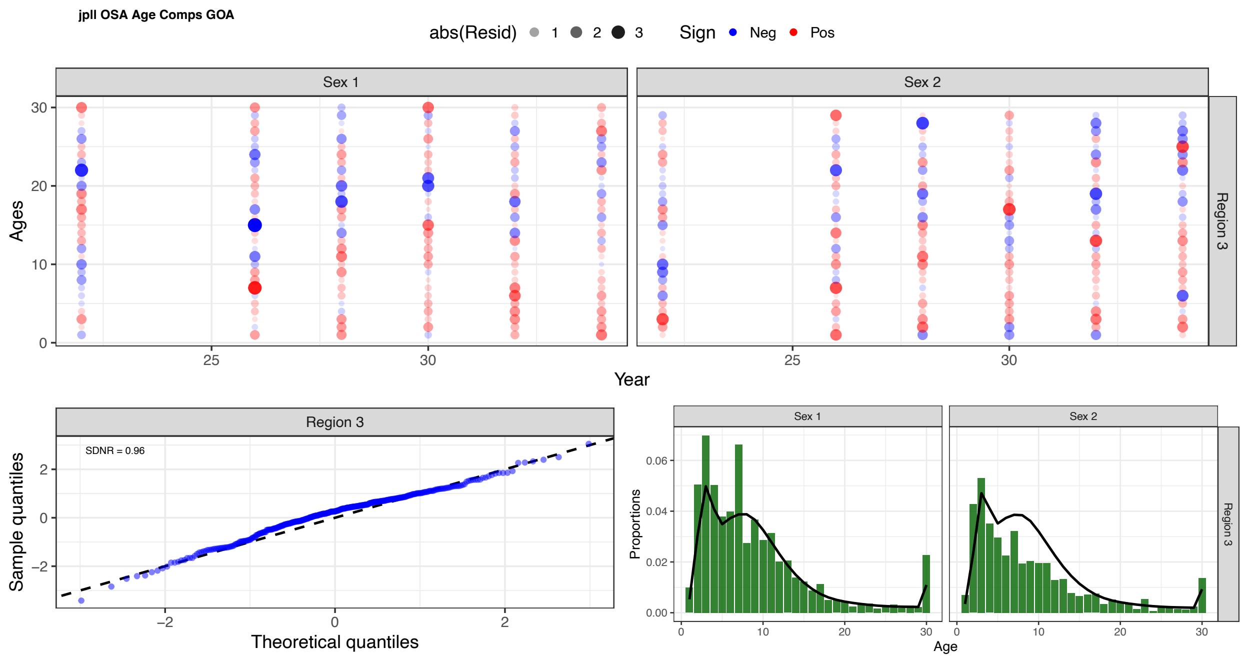

Figure 3D.14. Age composition fits from the three-region model for the Japanese longline survey in the Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

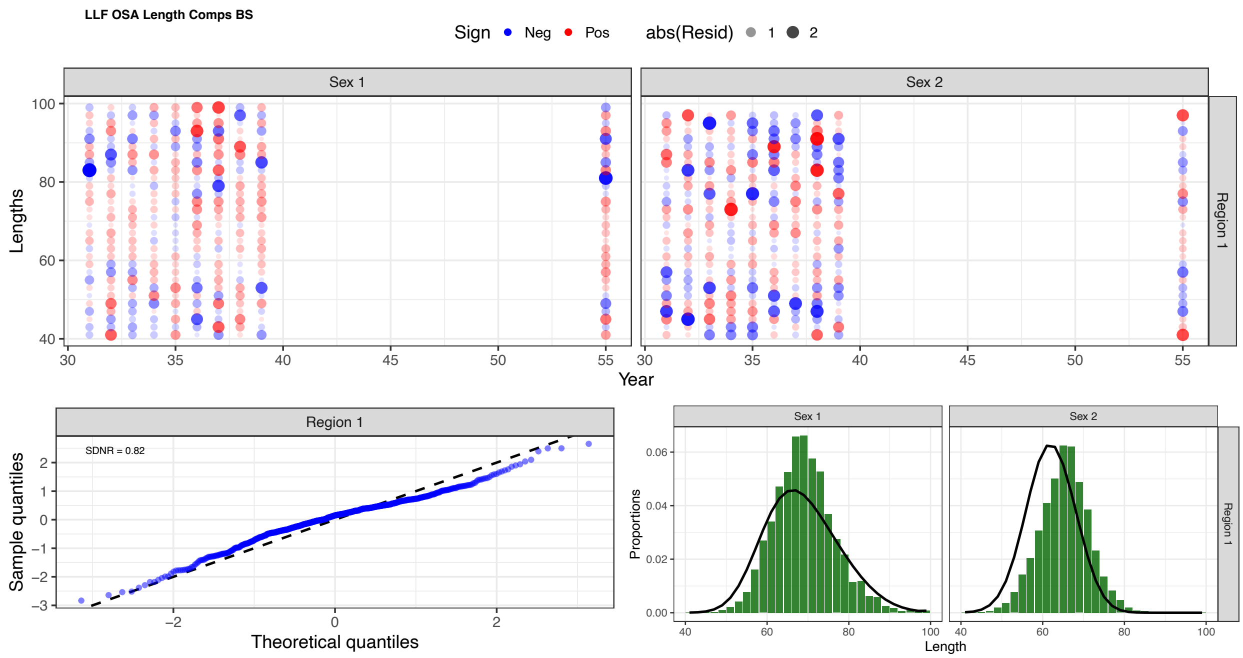

Figure 3D.15. Length composition fits from the three-region model for the longline fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

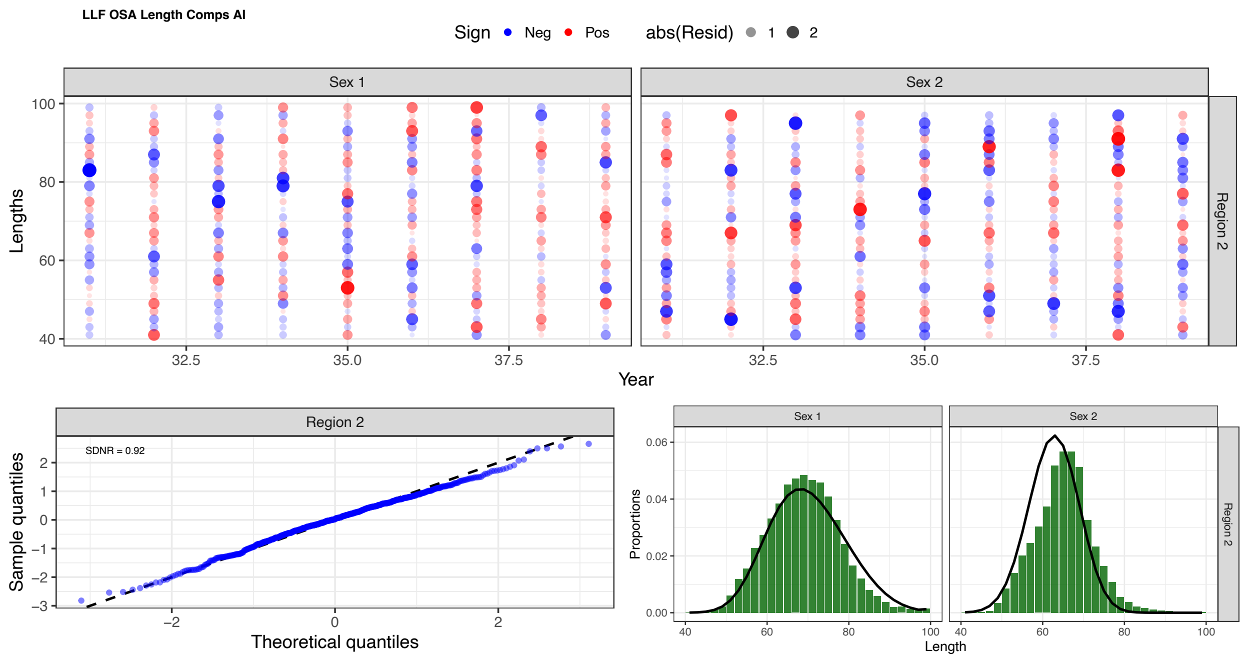

Figure 3D.16. Length composition fits from the three-region model for the longline fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

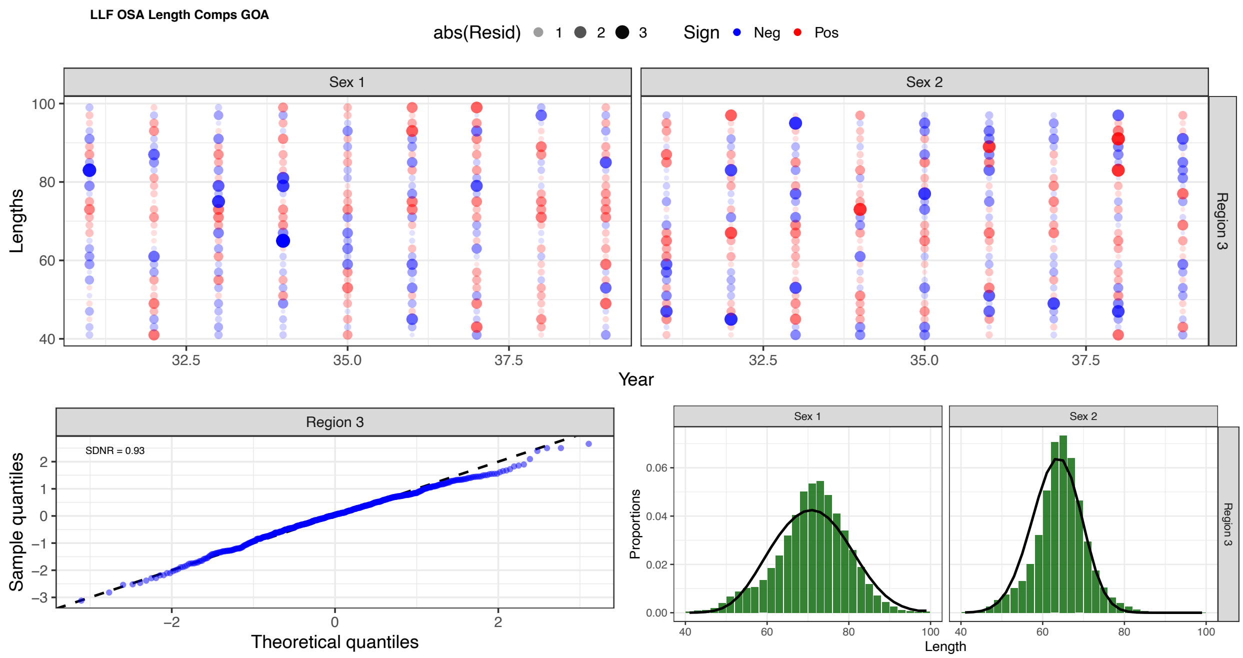

Figure 3D.17. Length composition fits from the three-region model for the longline fishery in the Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

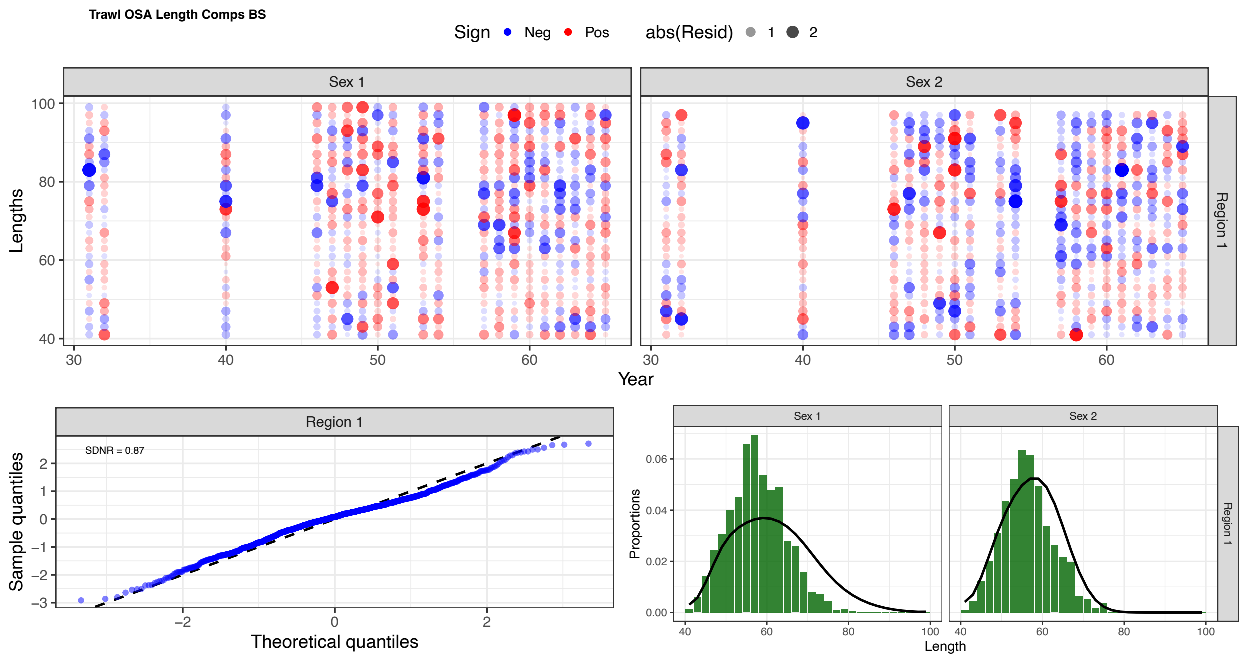

Figure 3D.18. Length composition fits from the three-region model for the trawl fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

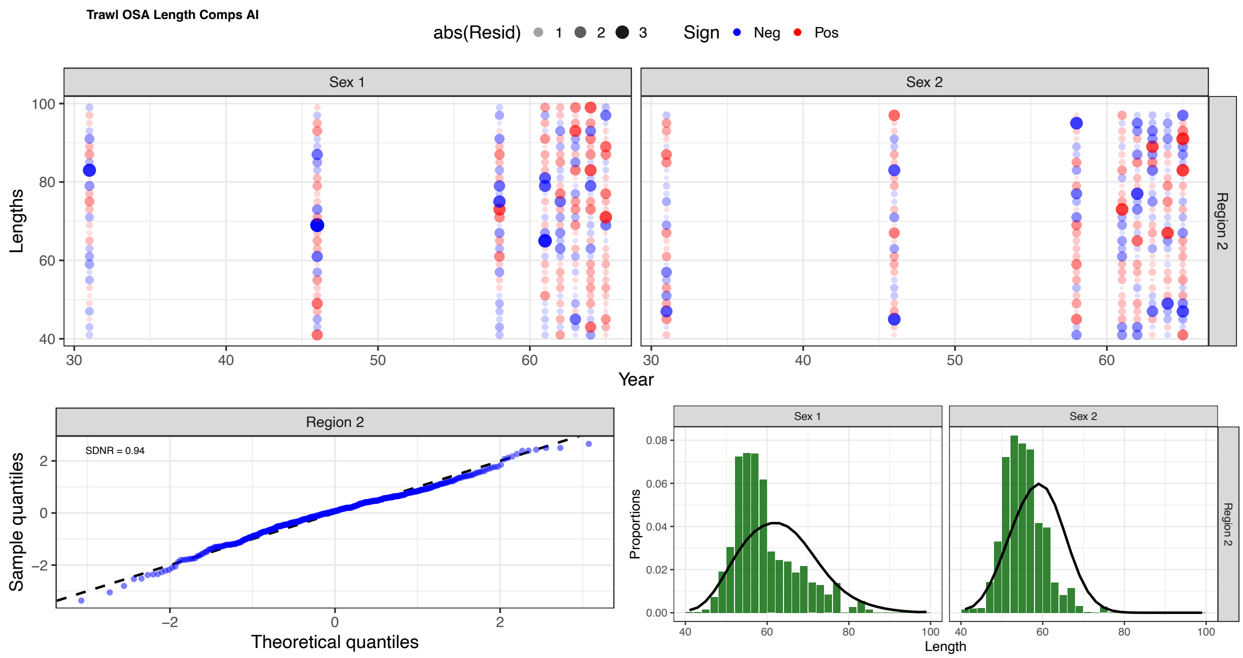

Figure 3D.19. Length composition fits from the three-region model for the trawl fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

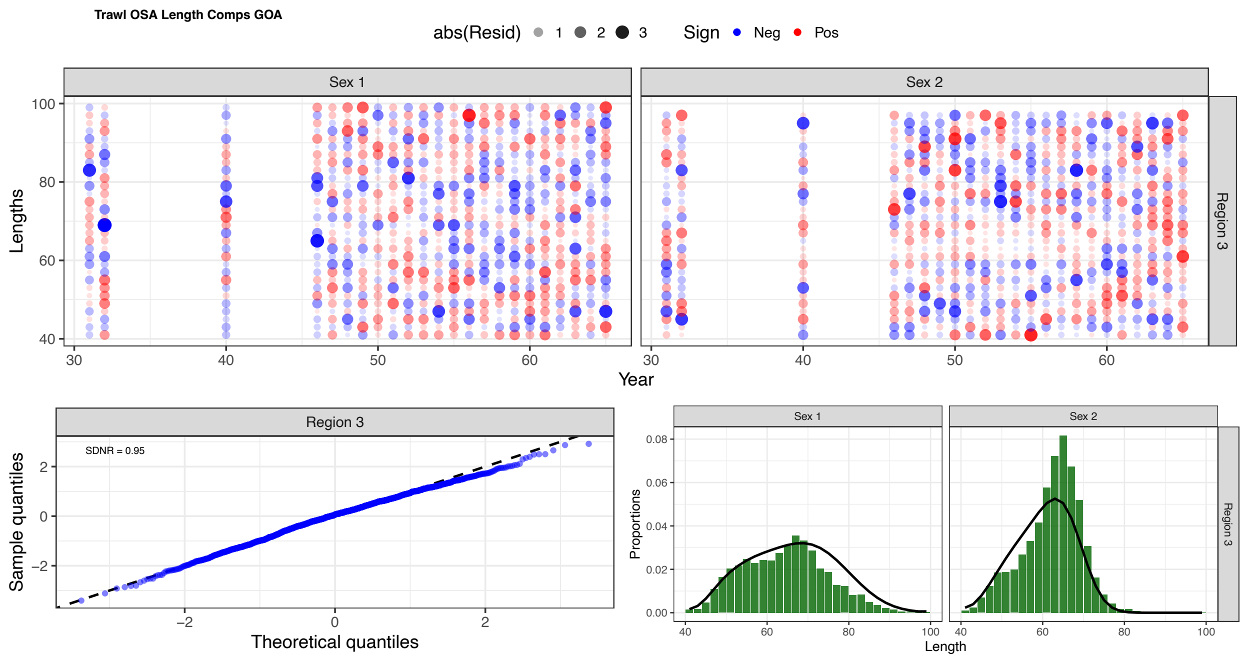

Figure 3D.20. Length composition fits from the three-region model for the trawl fishery in the Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

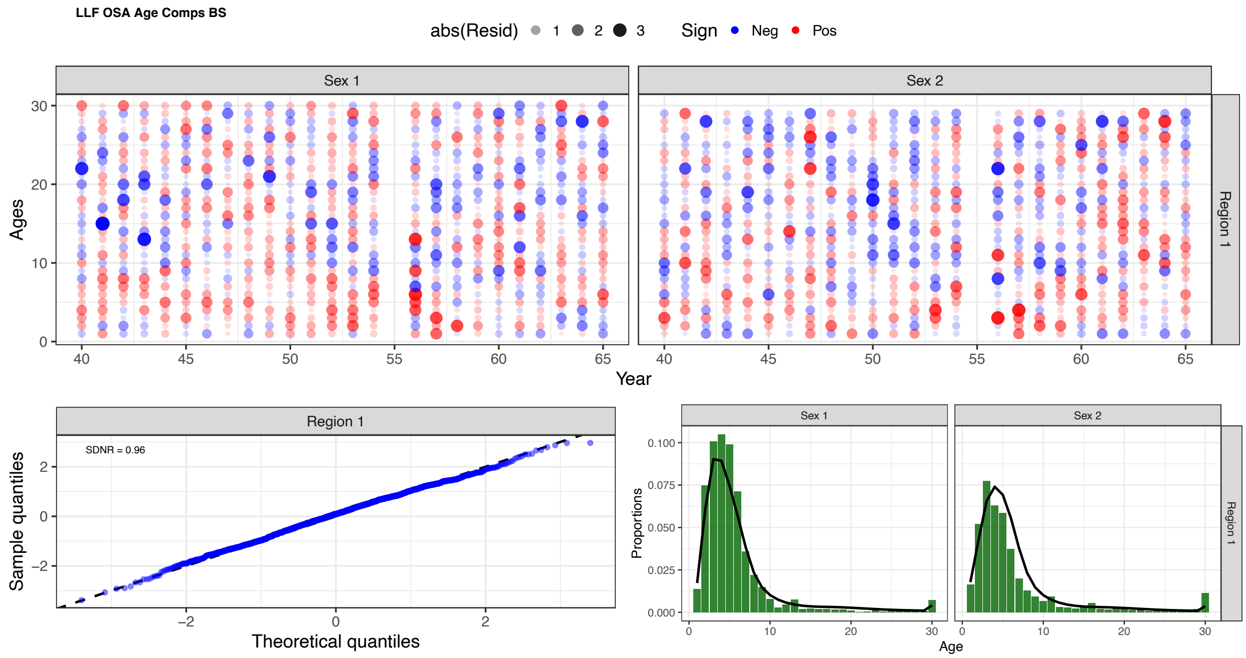

Figure 3D.21. Age composition fits from the five-region model for the longline fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

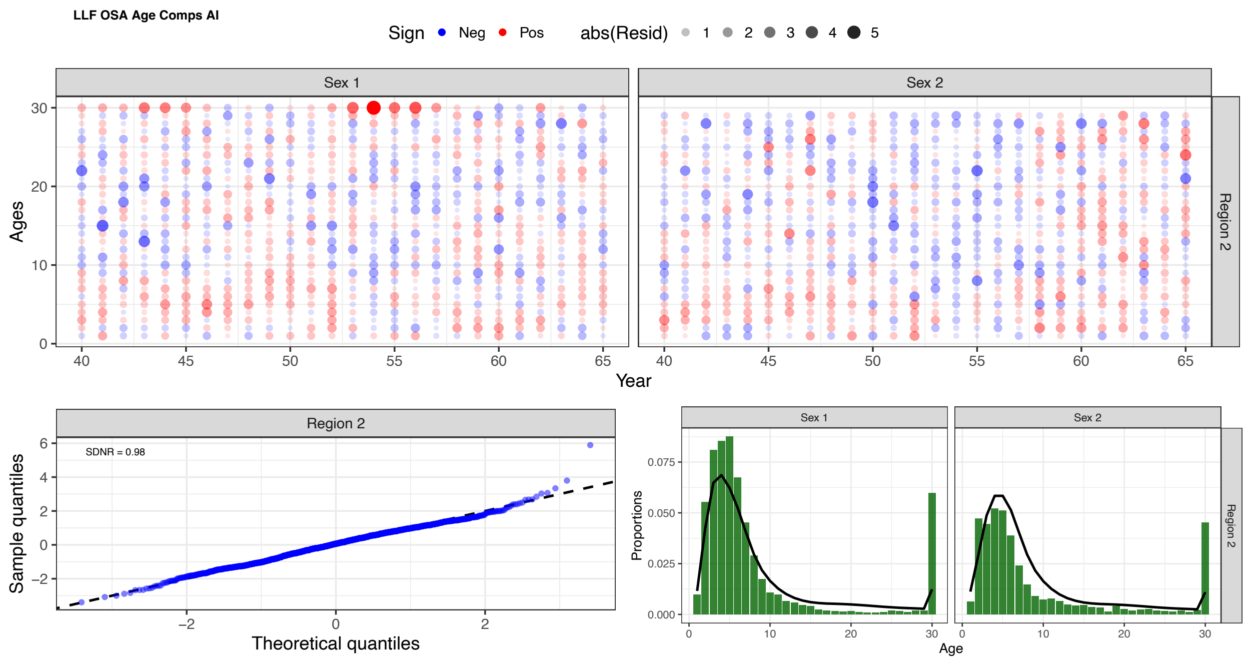

Figure 3D.22. Age composition fits from the five-region model for the longline fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

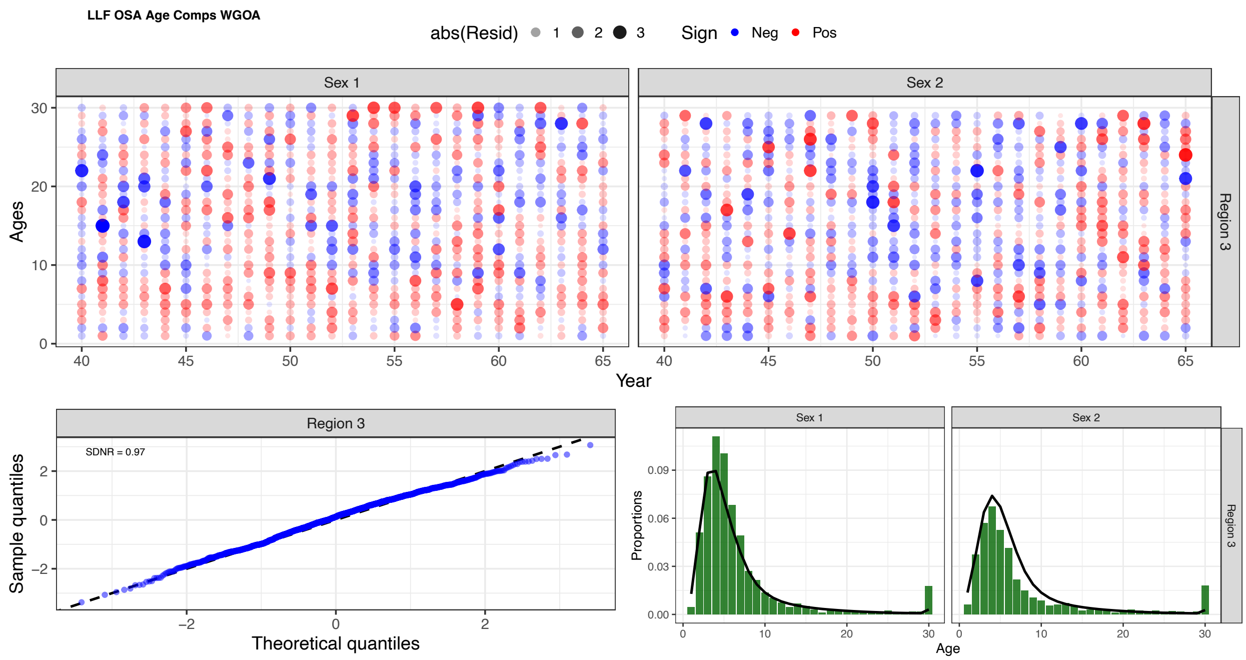

Figure 3D.23. Age composition fits from the five-region model for the longline fishery in the western Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

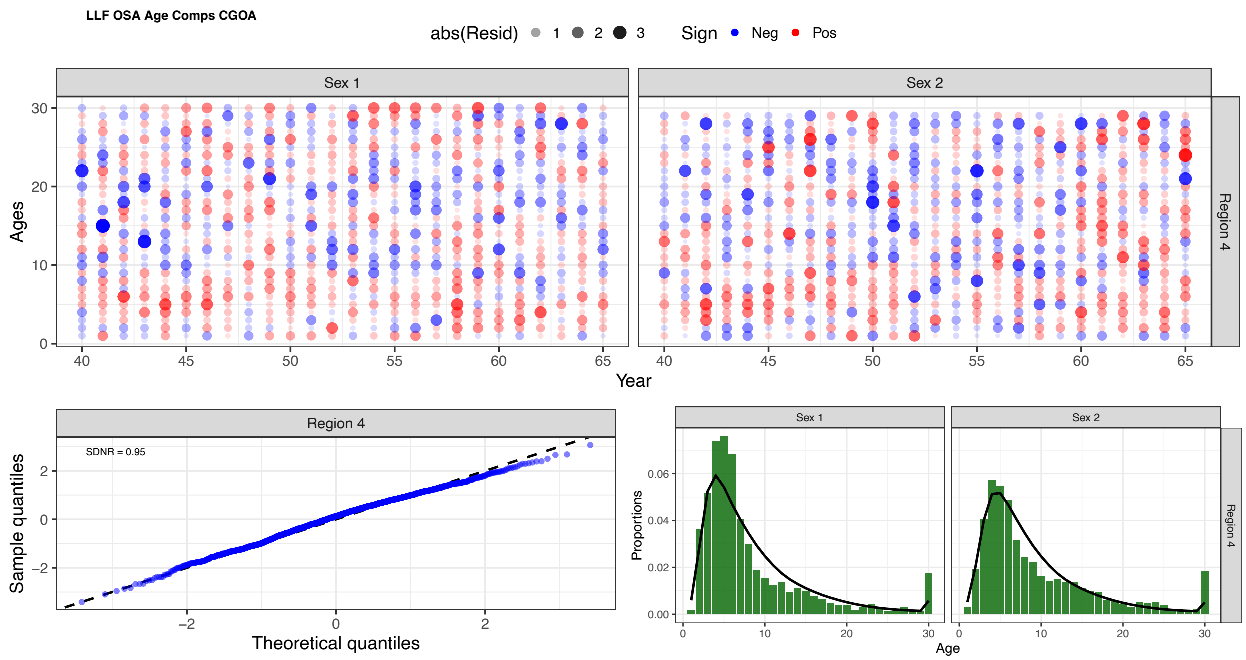

Figure 3D.24. Age composition fits from the five-region model for the longline fishery in the central Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

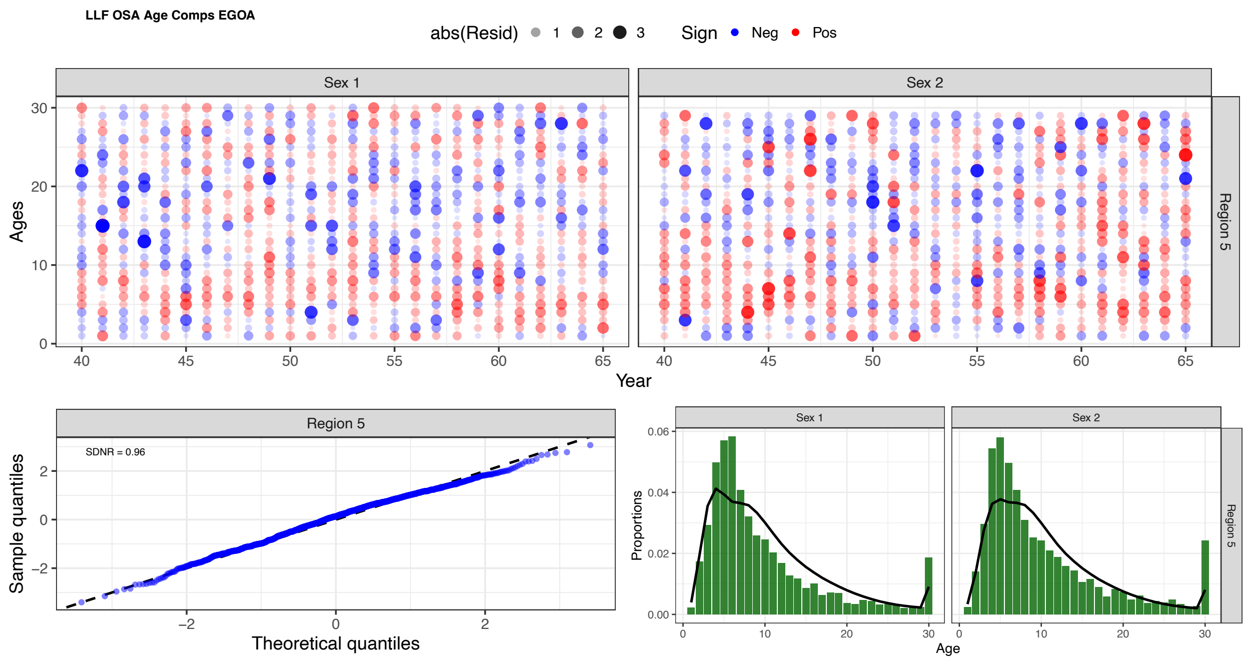

Figure 3D.25. Age composition fits from the five-region model for the longline fishery in the eastern Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

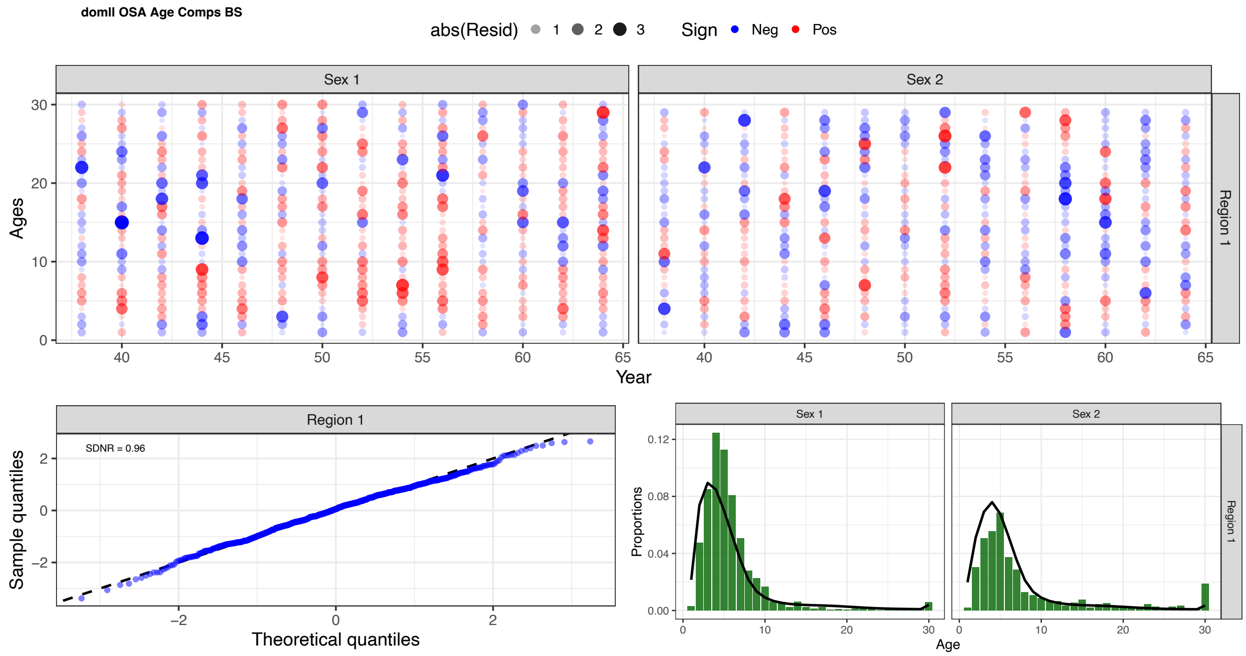

Figure 3D.24. Age composition fits from the five-region model for the domestic longline survey in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

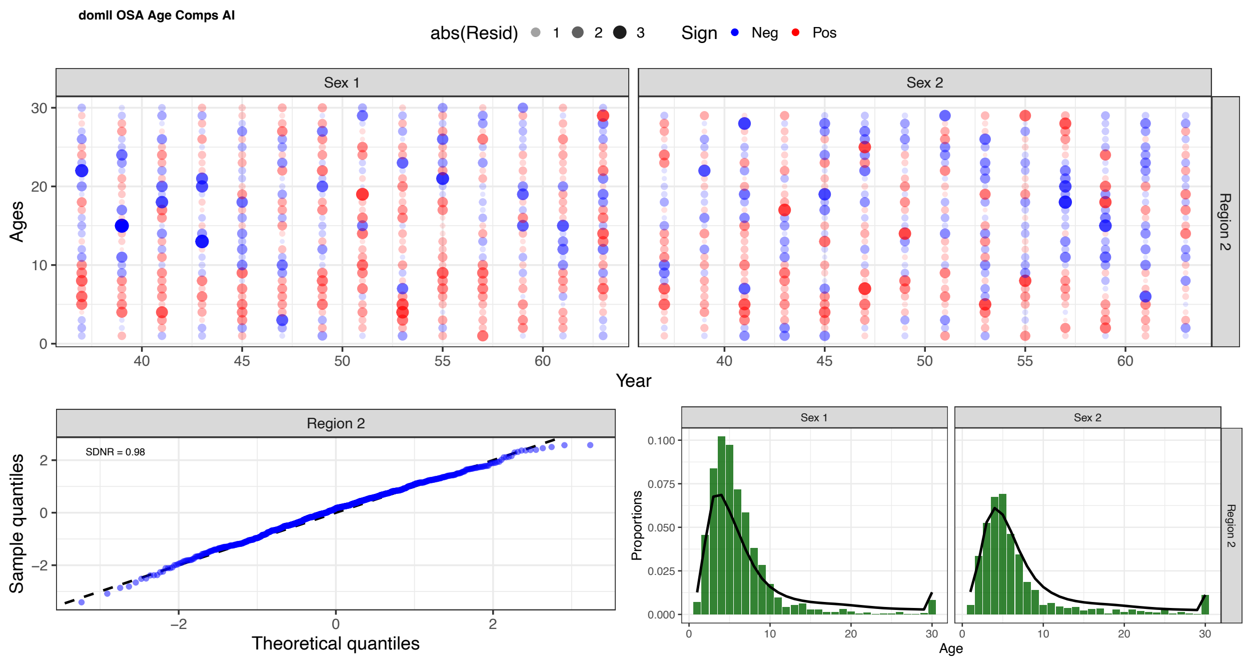

Figure 3D.25. Age composition fits from the five-region model for the domestic longline survey in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

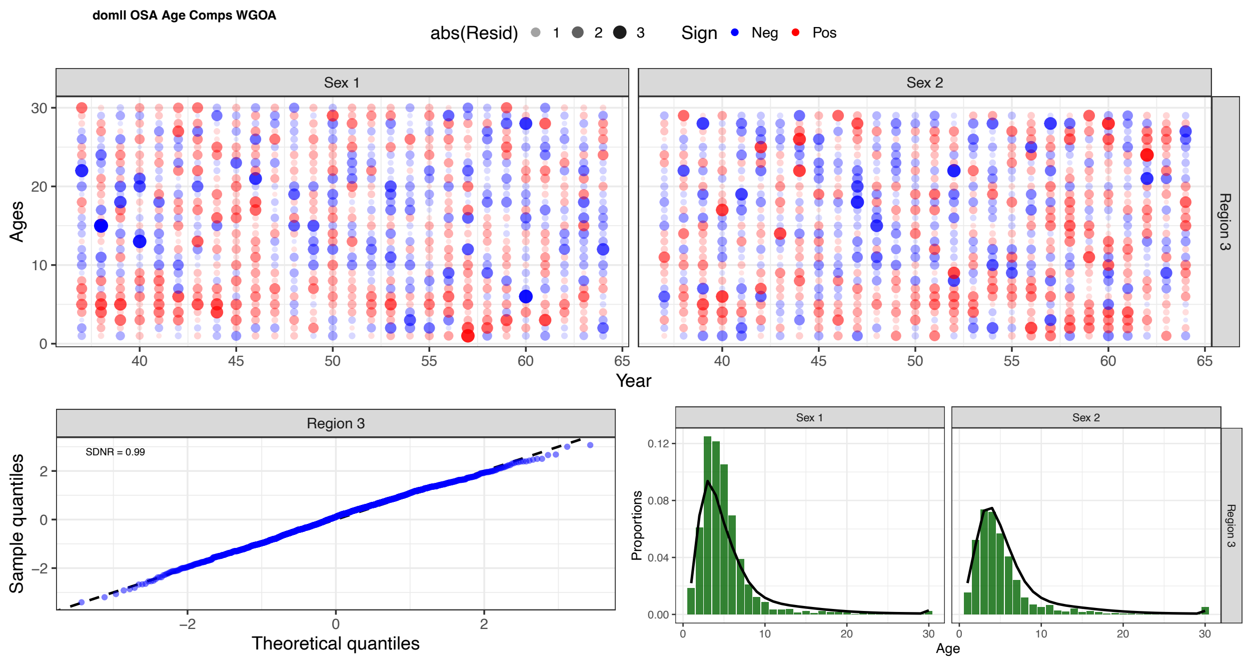

Figure 3D.26. Age composition fits from the five-region model for the domestic longline survey in the western Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

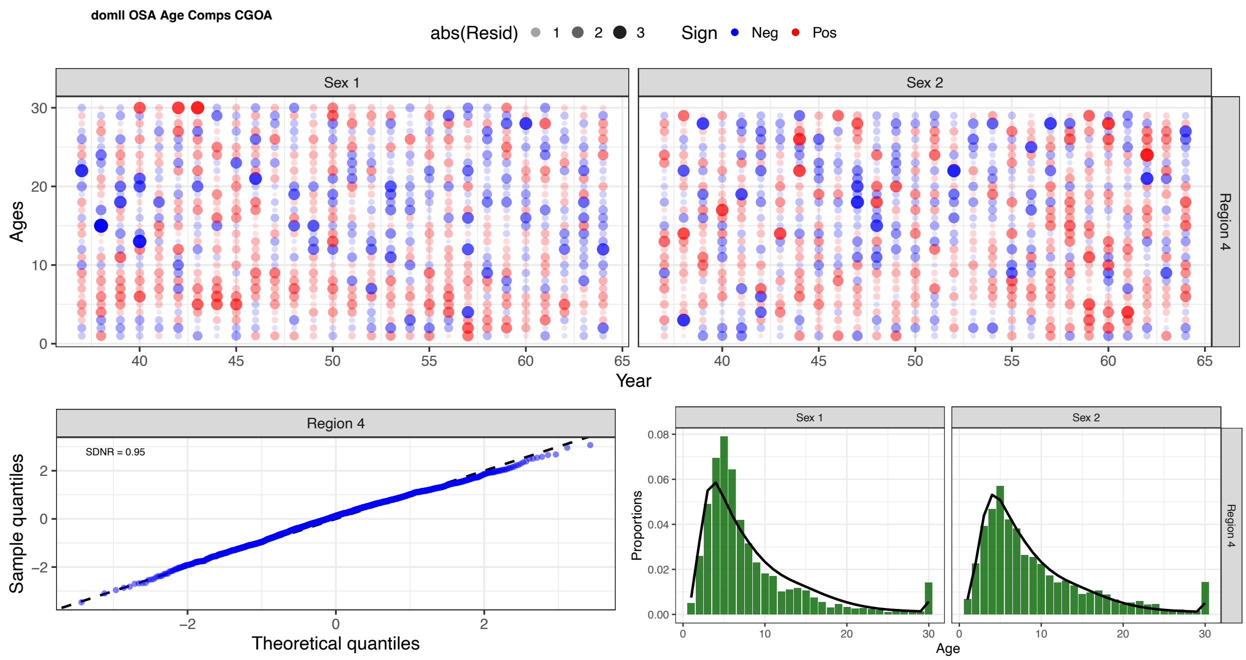

Figure 3D.27. Age composition fits from the five-region model for the domestic longline survey in the central Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

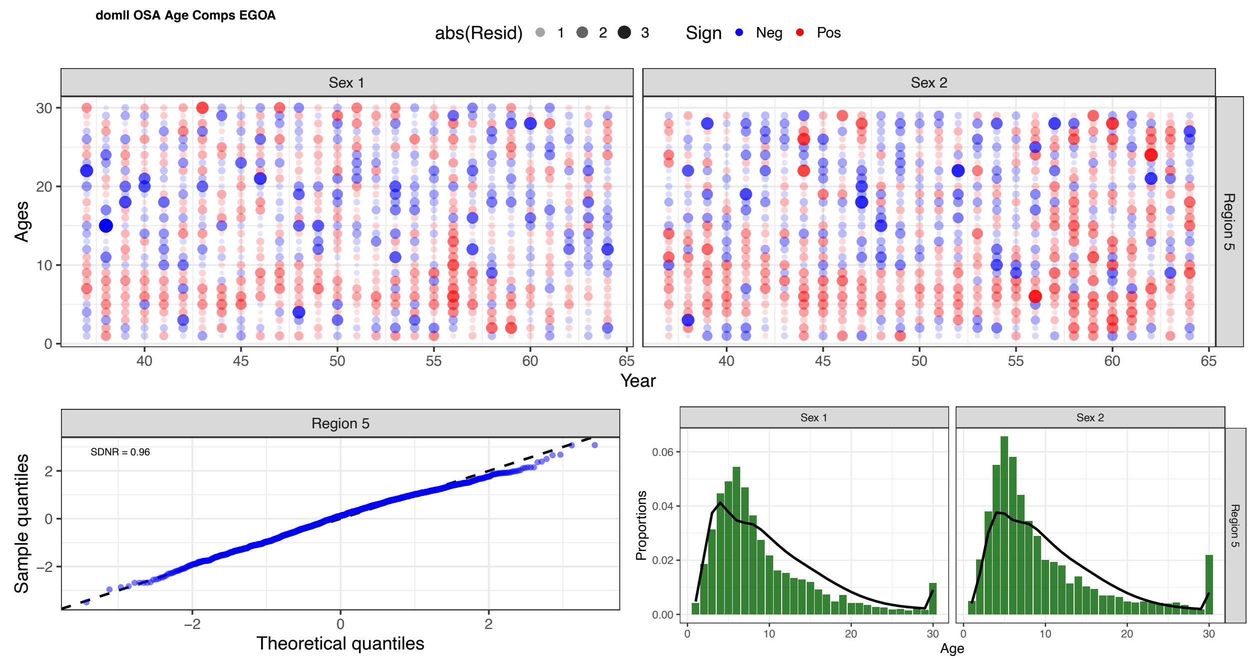

Figure 3D.28. Age composition fits from the five-region model for the domestic longline survey in the eastern Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

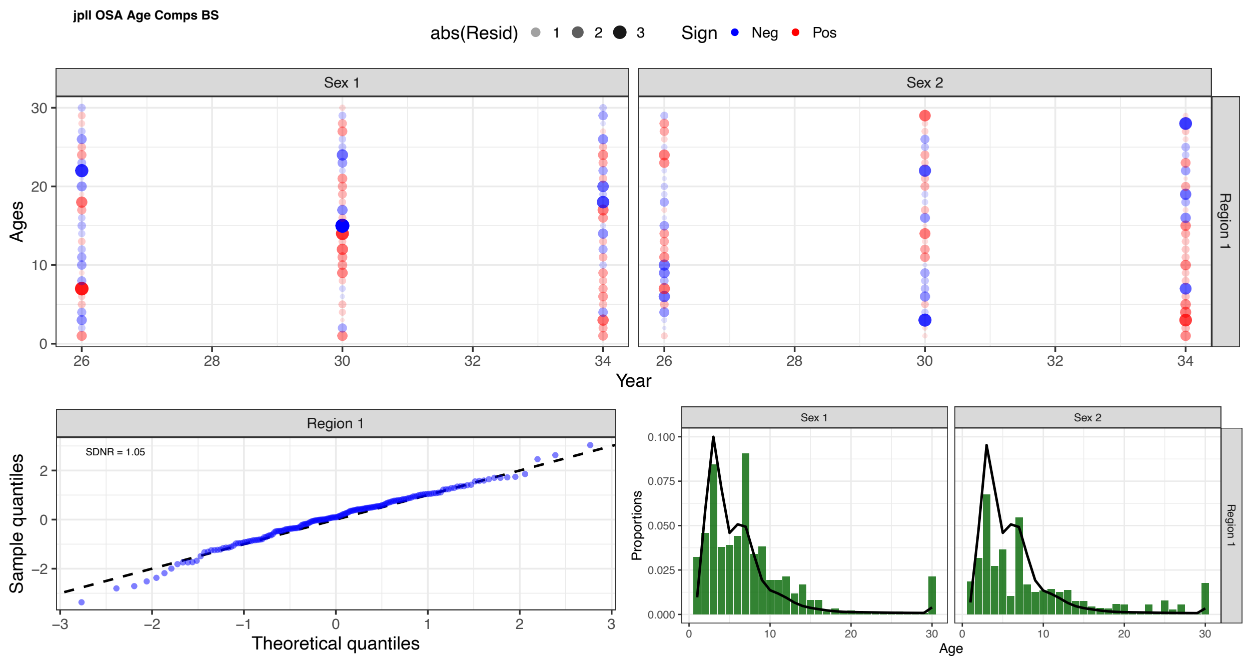

Figure 3D.29. Age composition fits from the five-region model for the Japanese longline survey in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

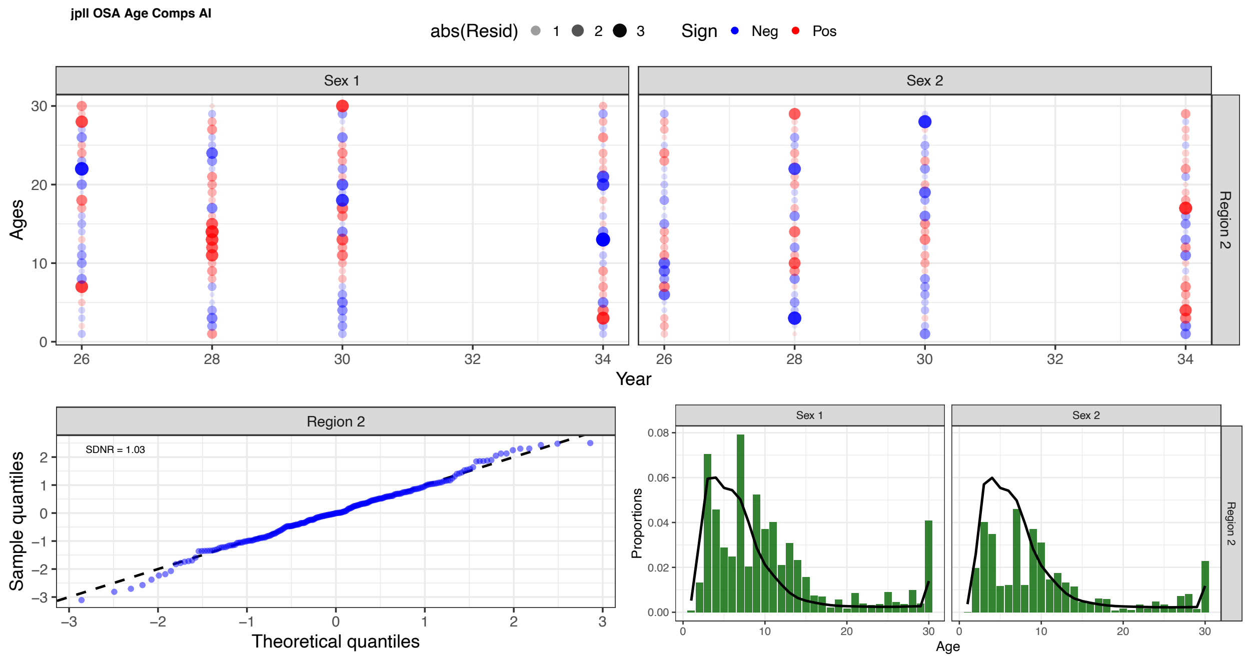

Figure 3D.30. Age composition fits from the five-region model for the Japanese longline survey in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

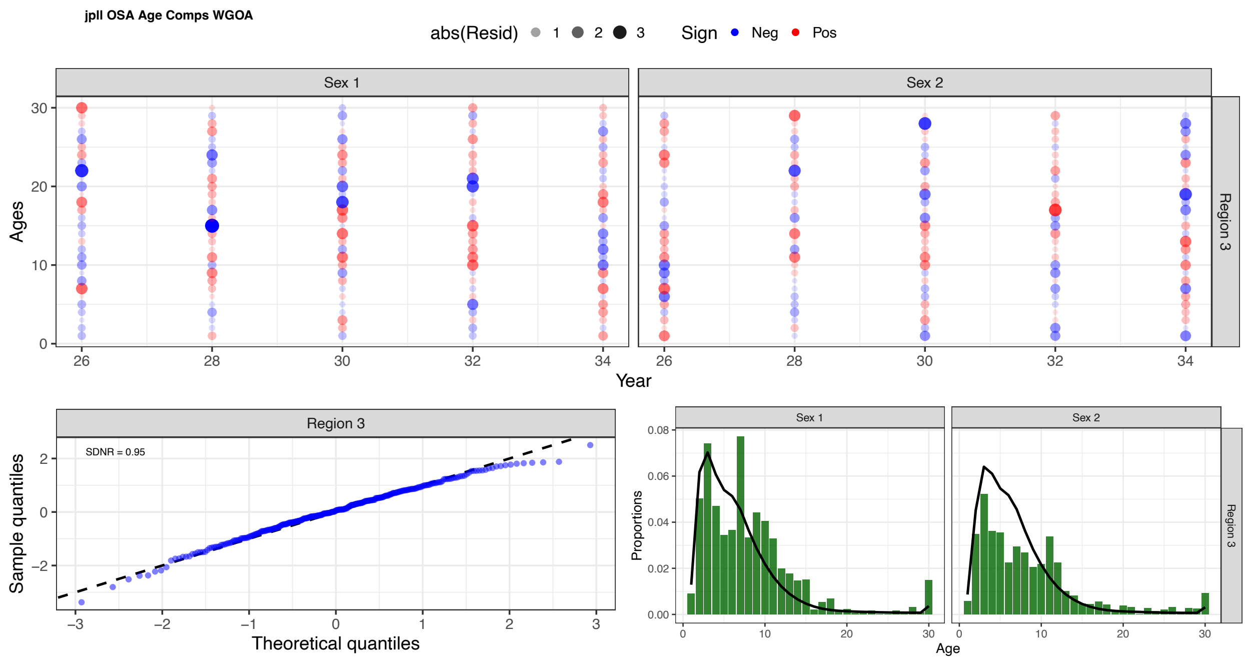

Figure 3D.31. Age composition fits from the five-region model for the Japanese longline survey in the western Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

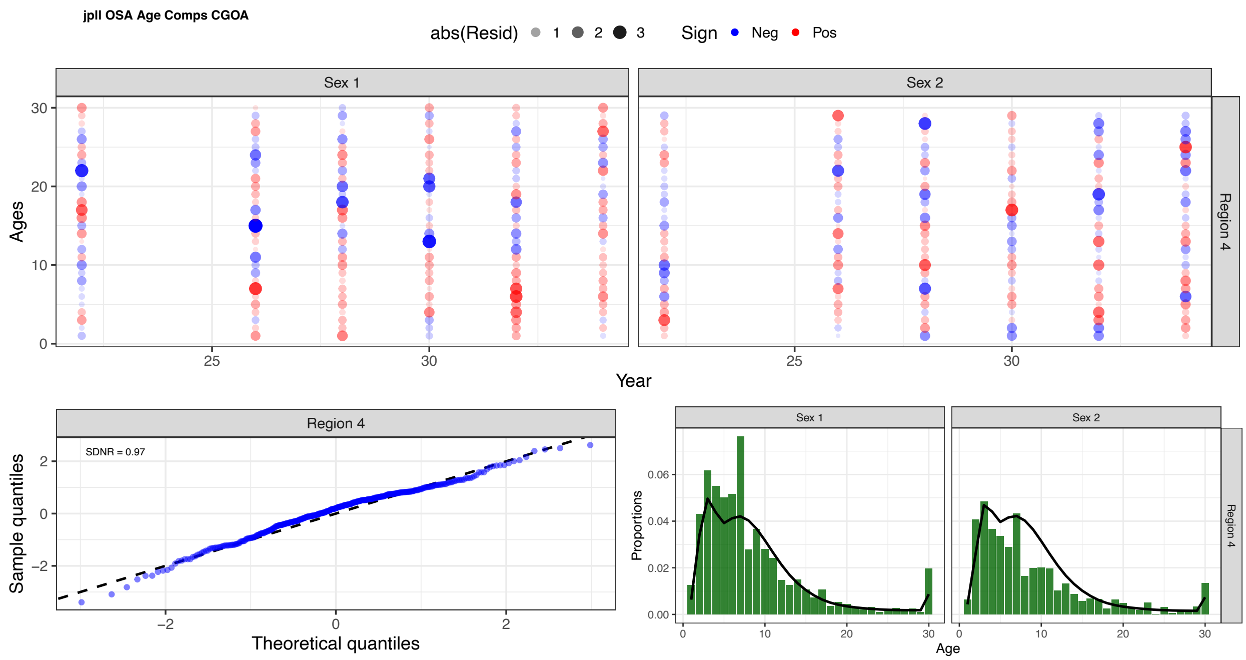

Figure 3D.32. Age composition fits from the five-region model for the Japanese longline survey in the central Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

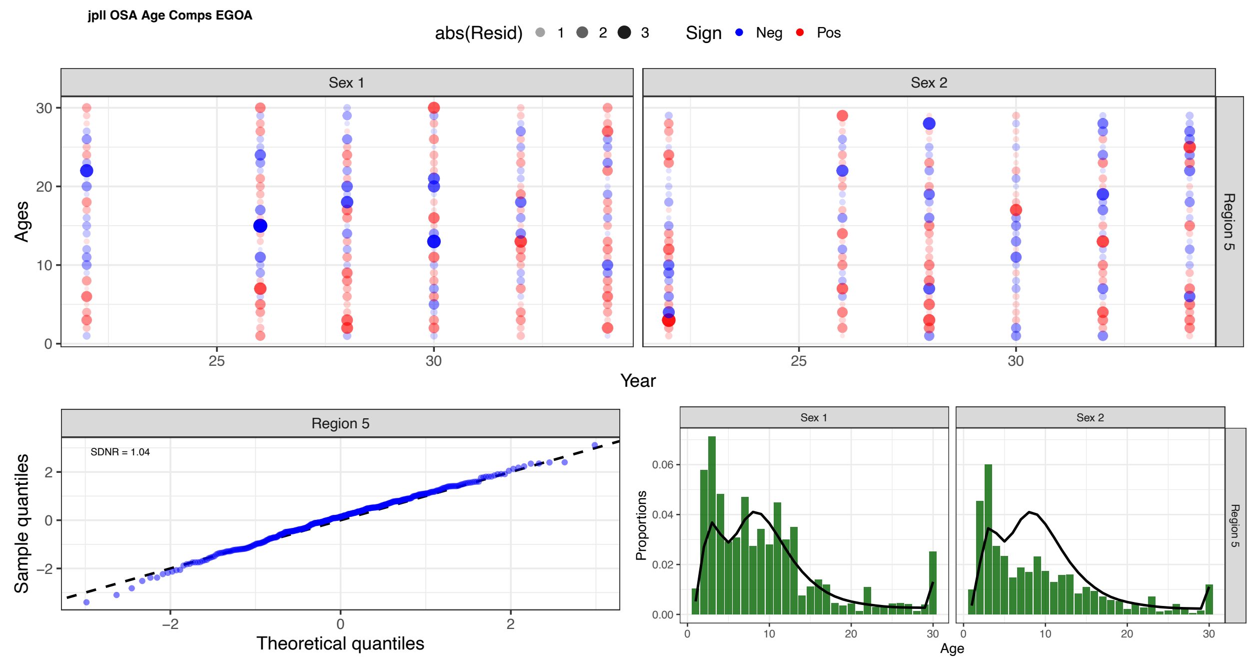

Figure 3D.33. Age composition fits from the five-region model for the Japanese longline survey in the eastern Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

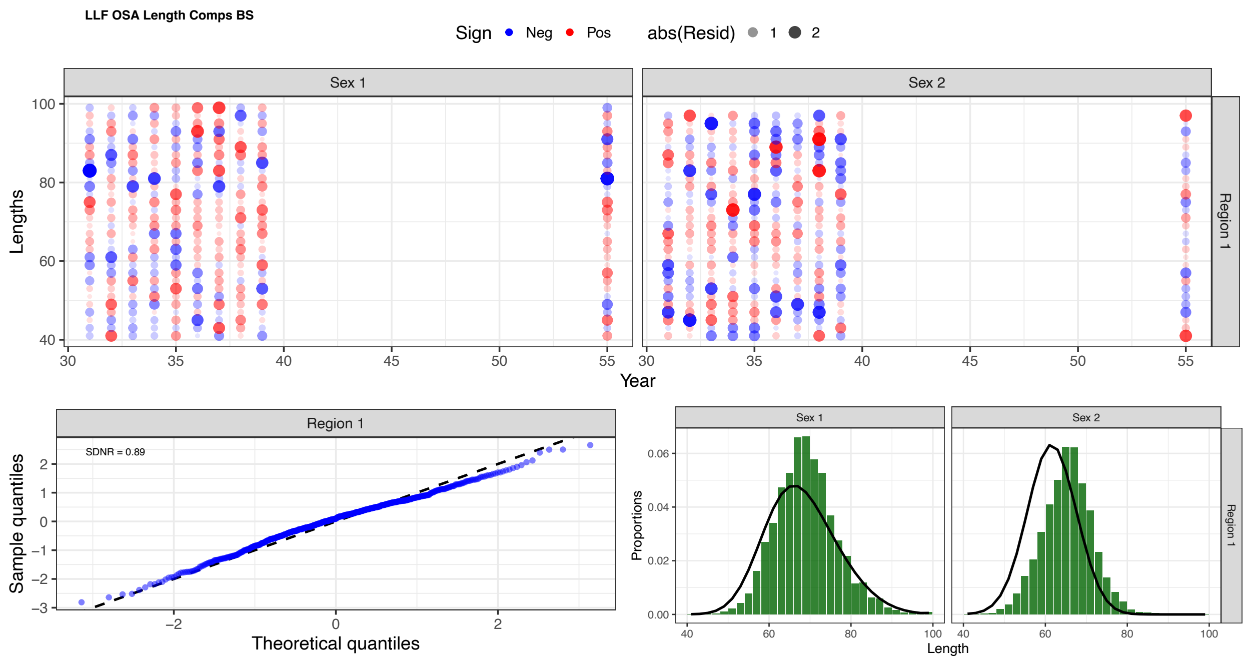

Figure 3D.34. Length composition fits from the five-region model for the longline fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

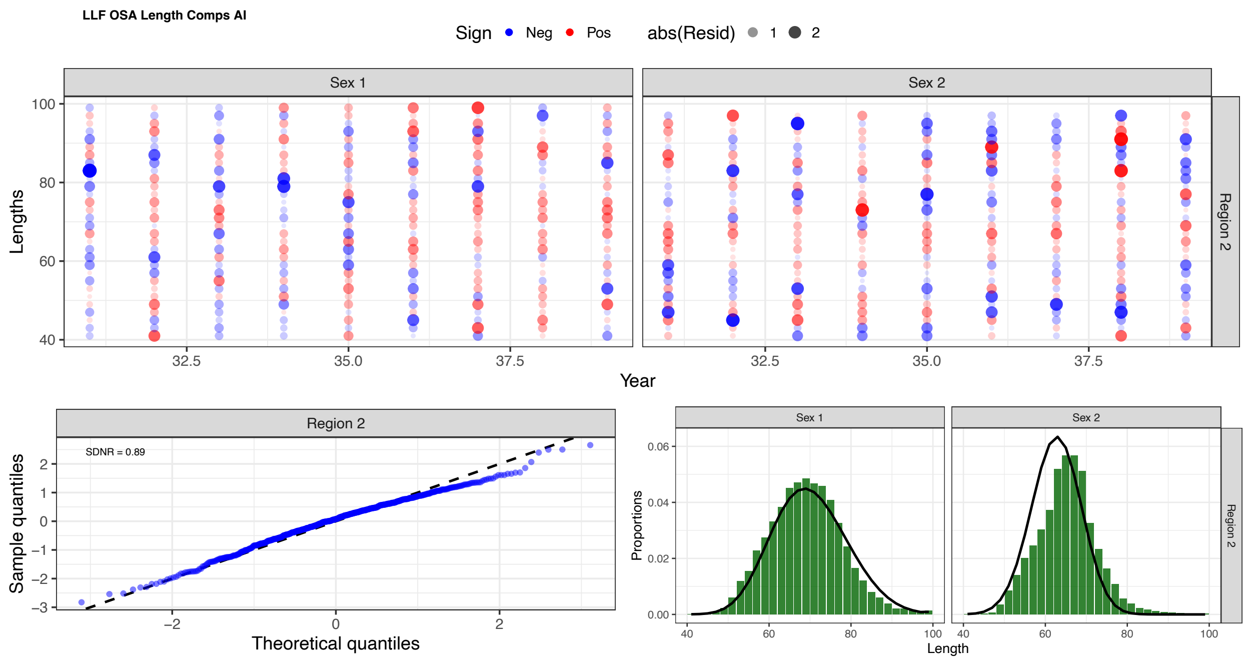

Figure 3D.35. Length composition fits from the five-region model for the longline fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

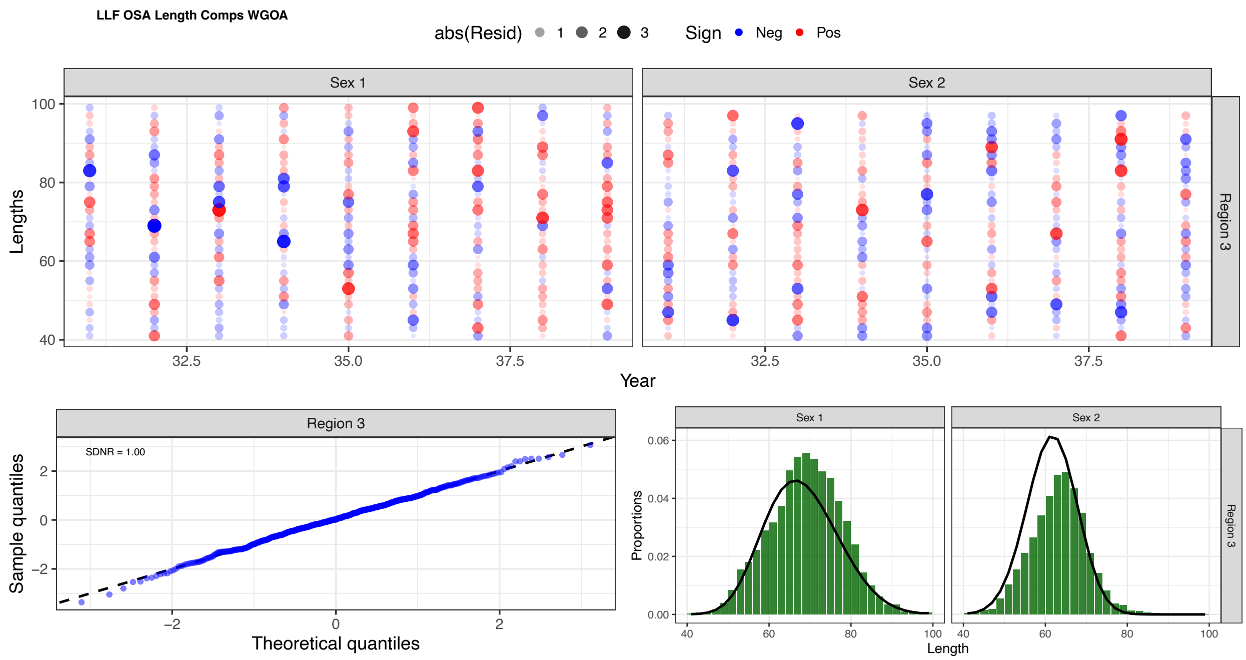

Figure 3D.36. Length composition fits from the five-region model for the longline fishery in the western Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

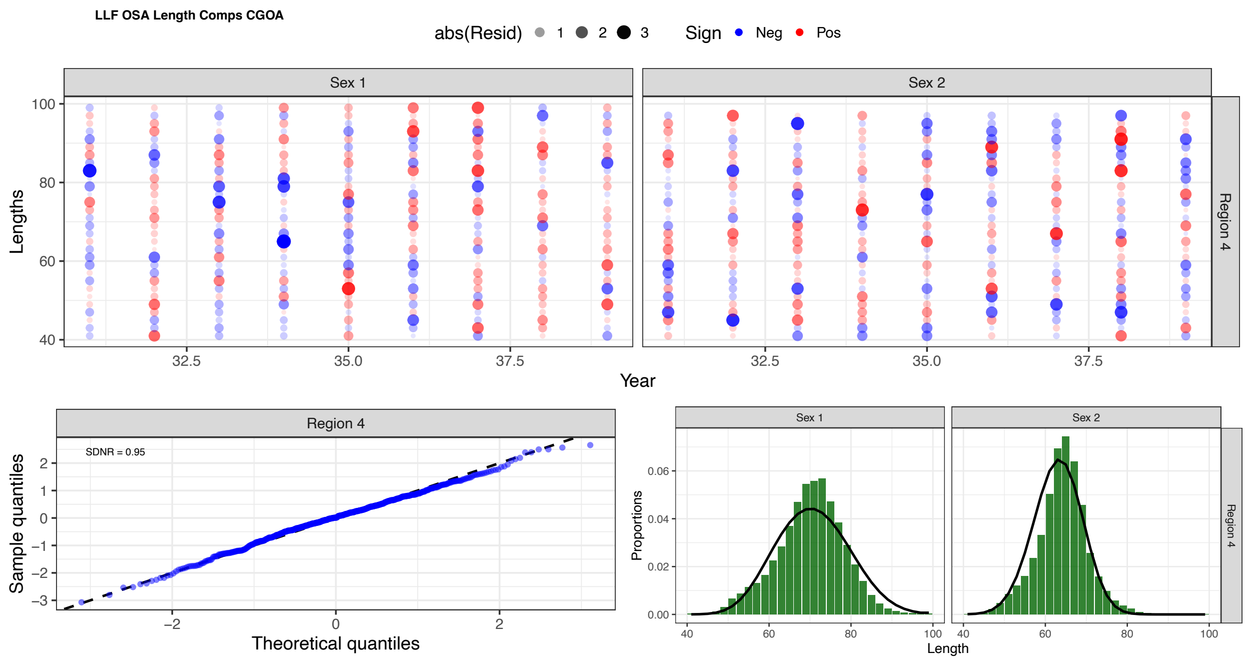

Figure 3D.37. Length composition fits from the five-region model for the longline fishery in the central Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

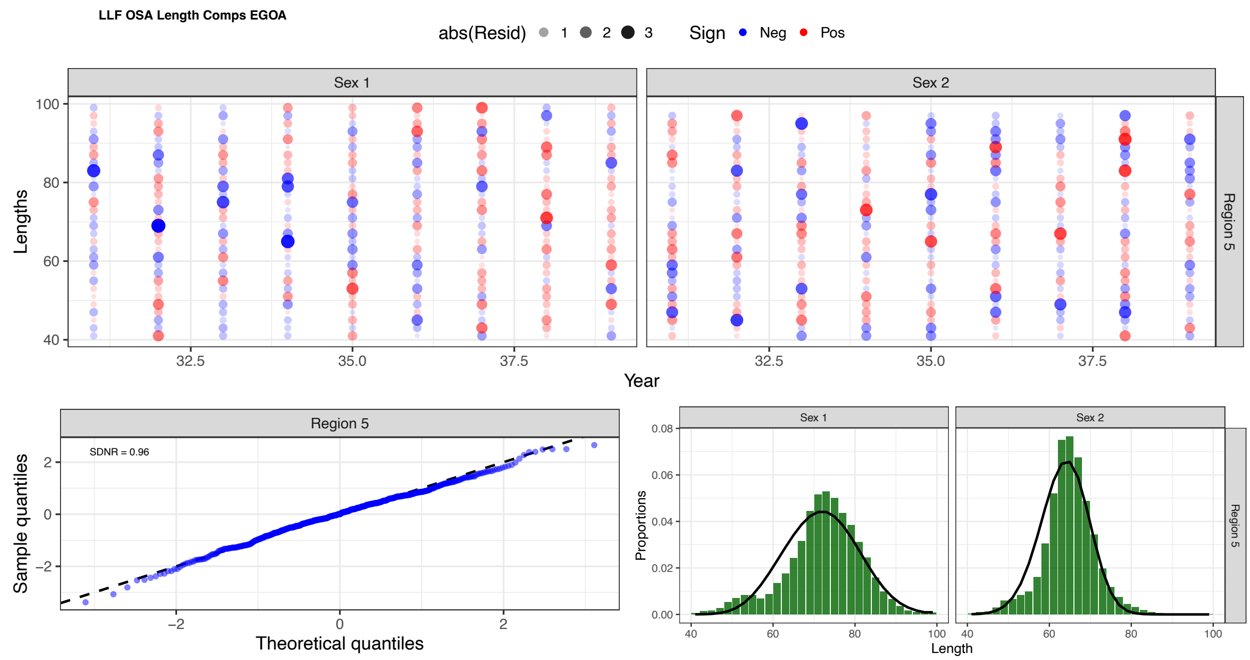

Figure 3D.38. Length composition fits from the five-region model for the longline fishery in the eastern Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

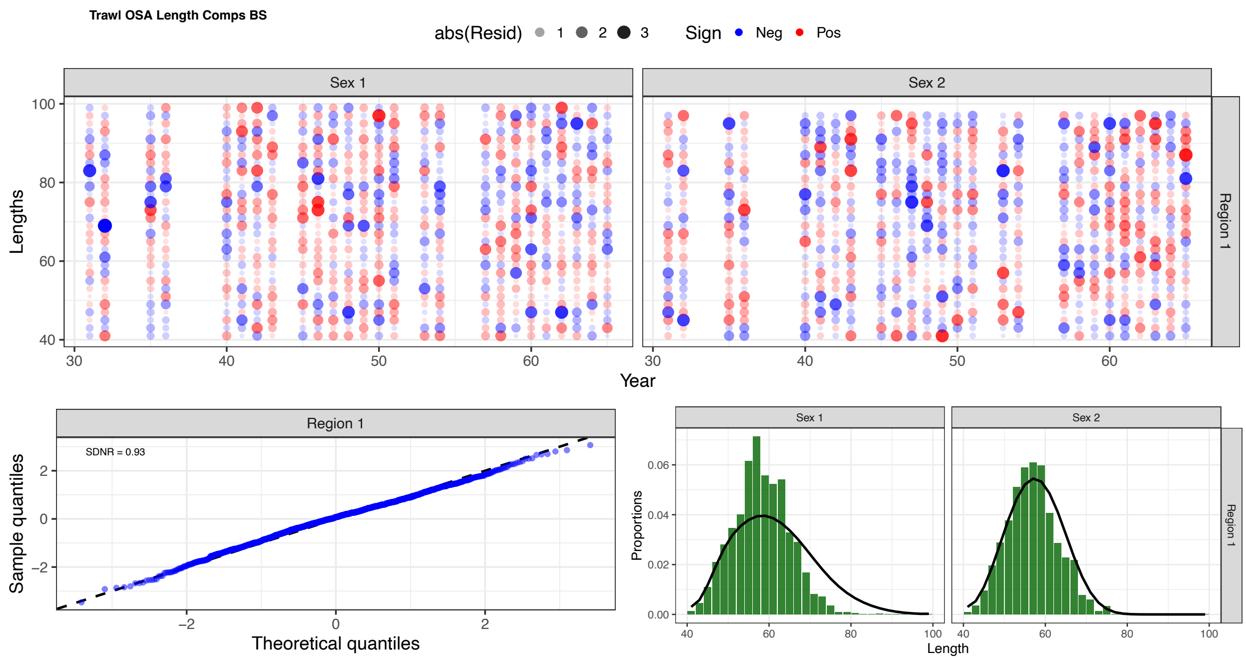

Figure 39. Length composition fits from the five-region model for the trawl fishery in the Bering Sea. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

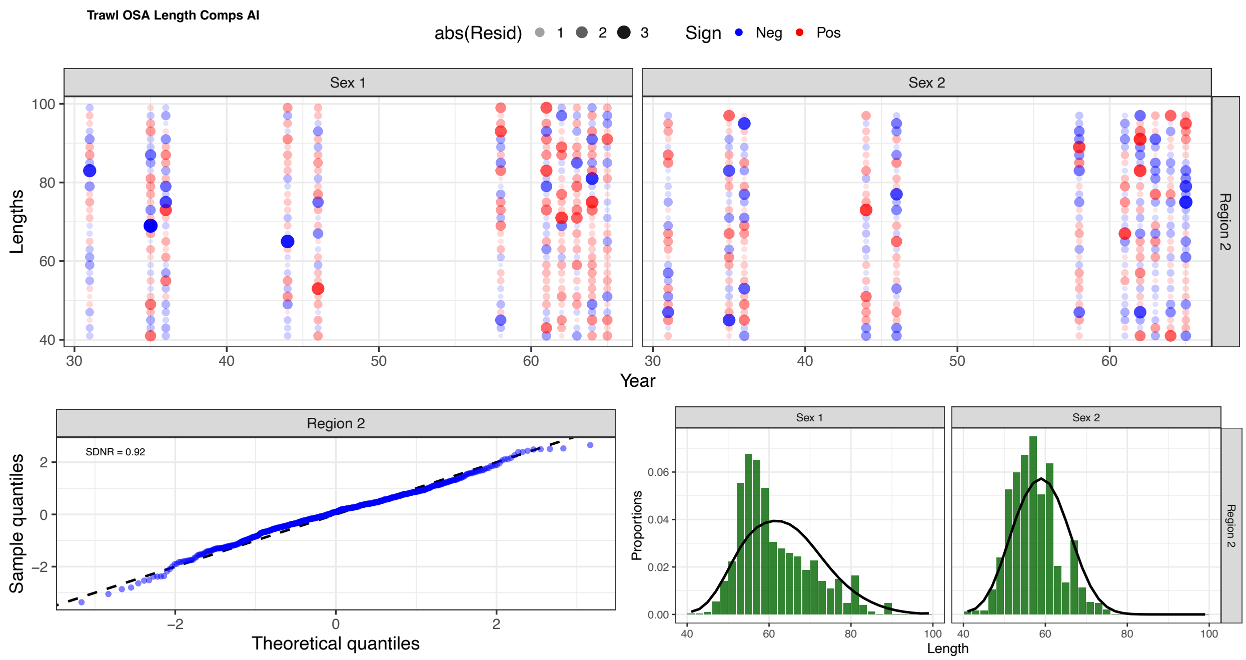

Figure 3D.40. Length composition fits from the five-region model for the trawl fishery in the Aleutian Islands. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

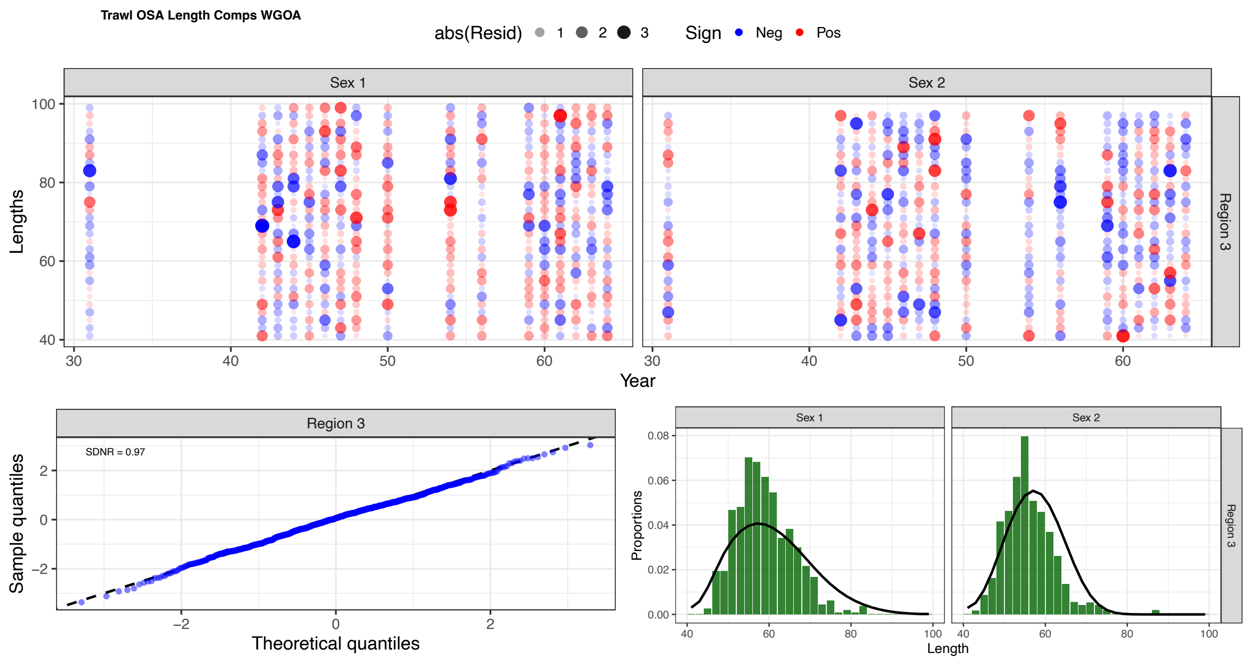

Figure 3D.41. Length composition fits from the five-region model for the trawl fishery in the western Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

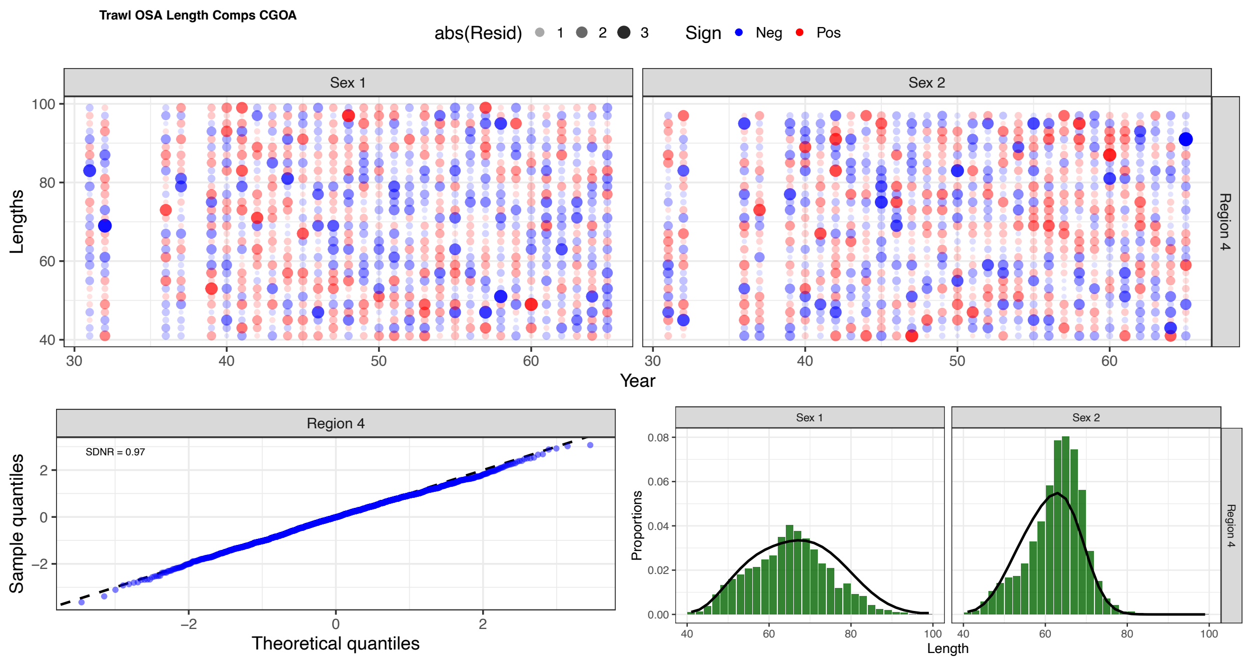

Figure 3D.42. Length composition fits from the five-region model for the trawl fishery in the central Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

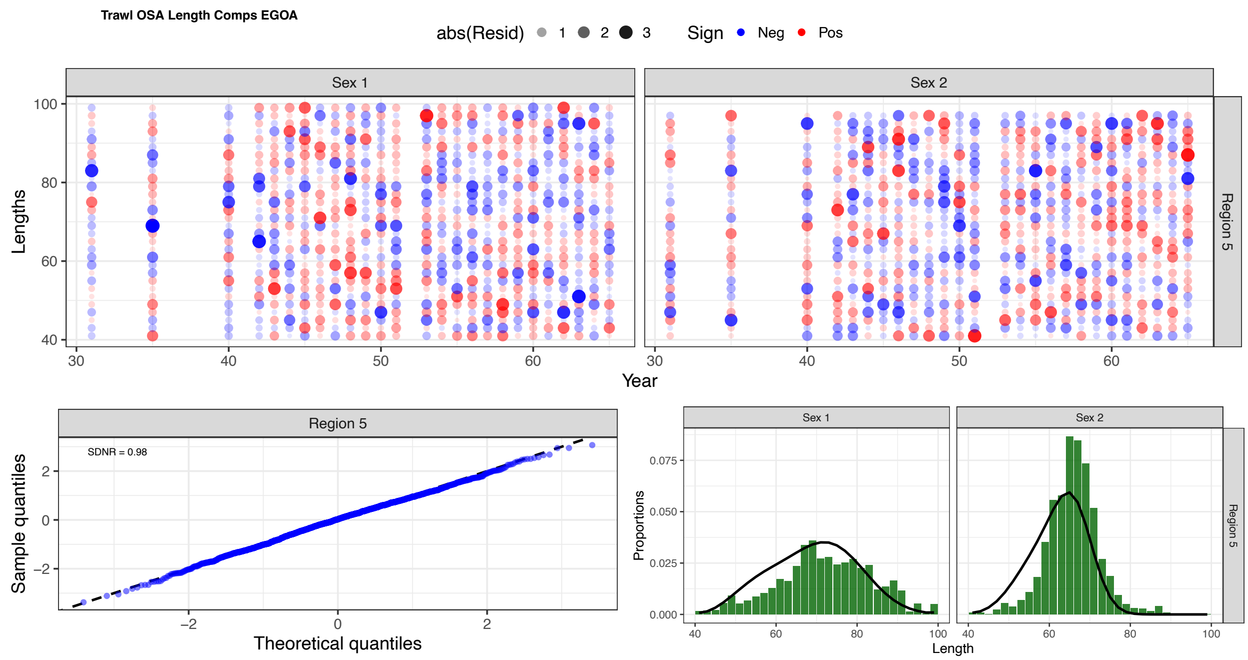

Figure 3D.43. Length composition fits from the five-region model for the trawl fishery in the eastern Gulf of Alaska. The upper panel depicts one-step ahead (OSA) residuals for females (Sex 1) and males (Sex 2). The bottom left panel represents a quantile-quantile plot of OSA residuals, with the standard deviation of the normalized residual displayed on the left corner. The bottom right panel illustrates aggregate fits to the composition data (green denotes observed, black denotes expected).

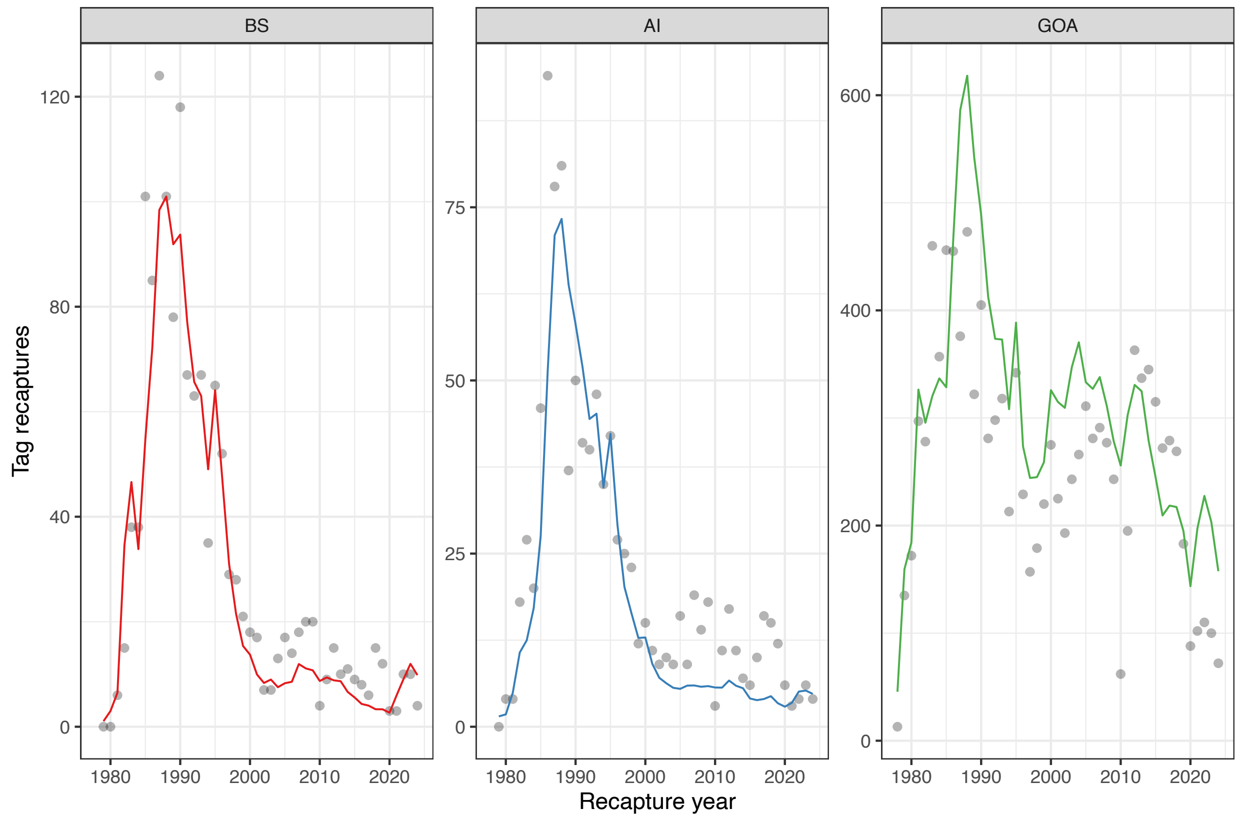

Figure 3D.44. Fits to tag recaptures across regions for the three-region model. Colored lines represent the model expectation while grey points denote observed recaptures.

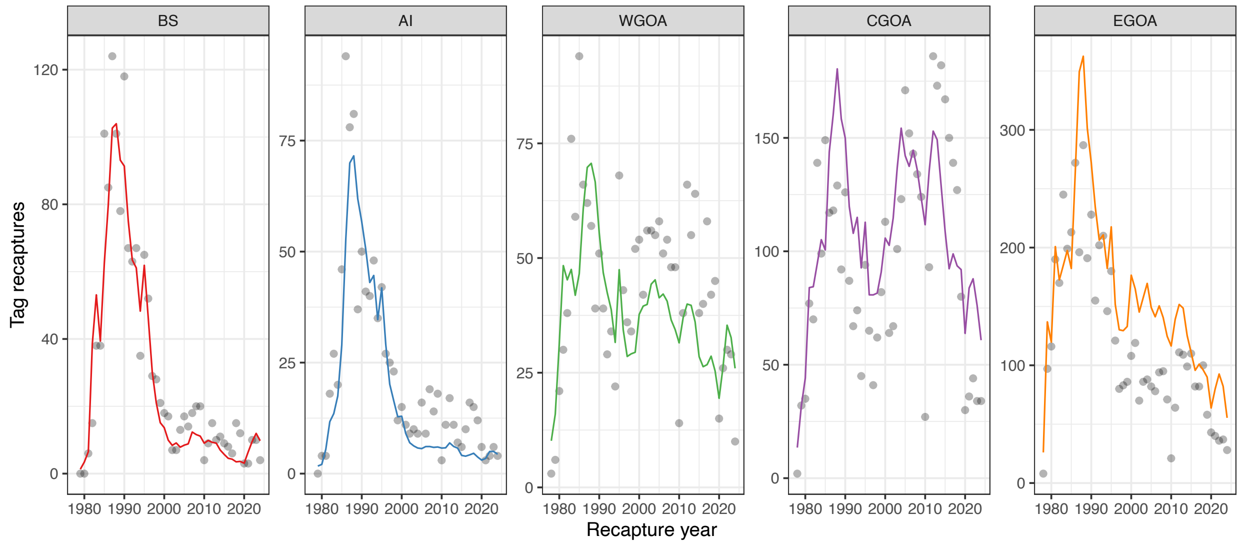

Figure 3D.45. Fits to tag recaptures across regions for the five-region model. Colored lines represent the model expectation while grey points denote observed recaptures.

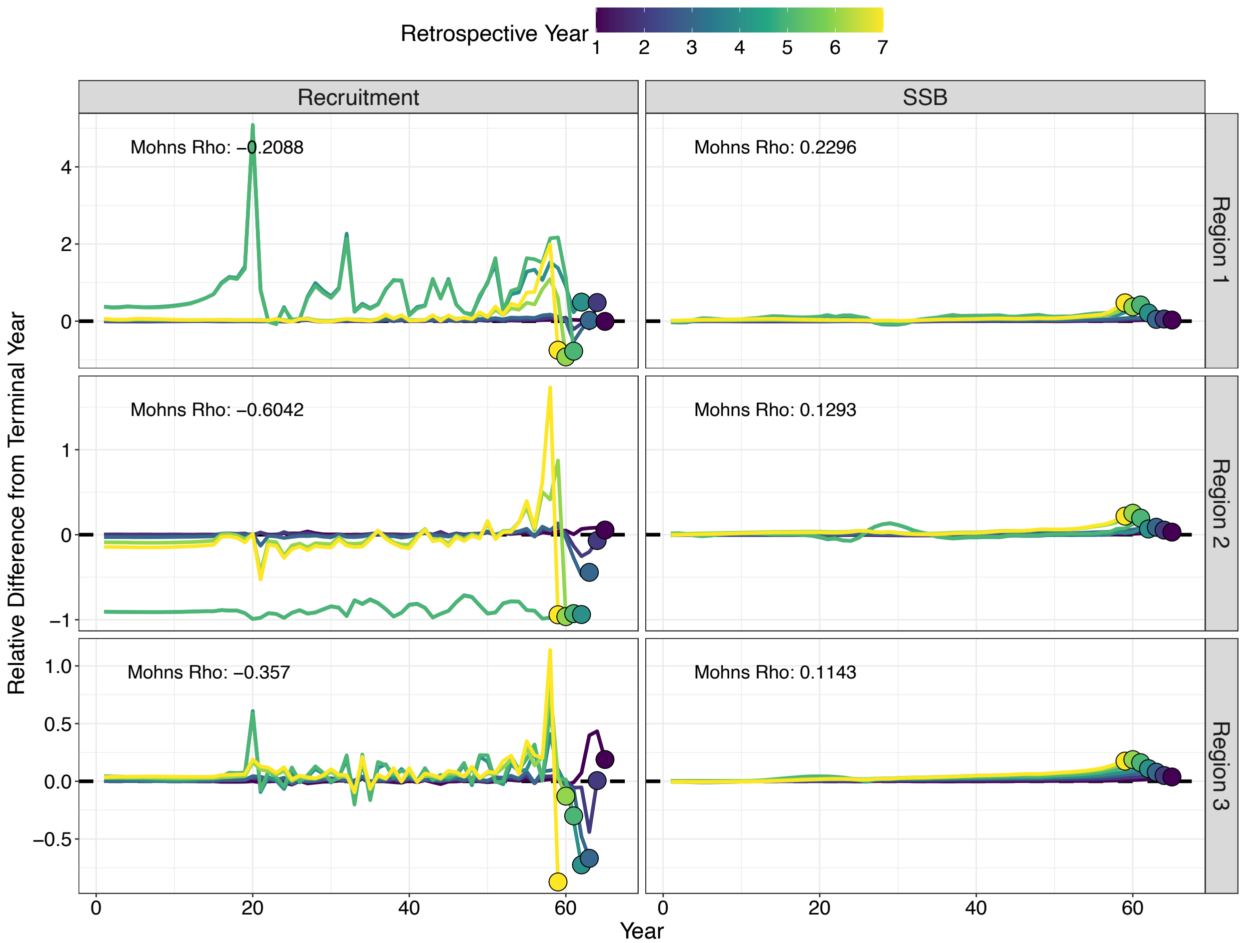

Figure 3D.46. Retrospective analysis from the three-region model. Column panels represent time series of recruitment and spawning stock biomass (*SSB*) as data are sequentially removed. Row panels denote the modeled regions, where region 1 represents the Bering Sea, region 2 represents the Aleutian Islands, and region 3 represents the Gulf of Alaska. Mohn's rho values are shown in the upper left of each panel.

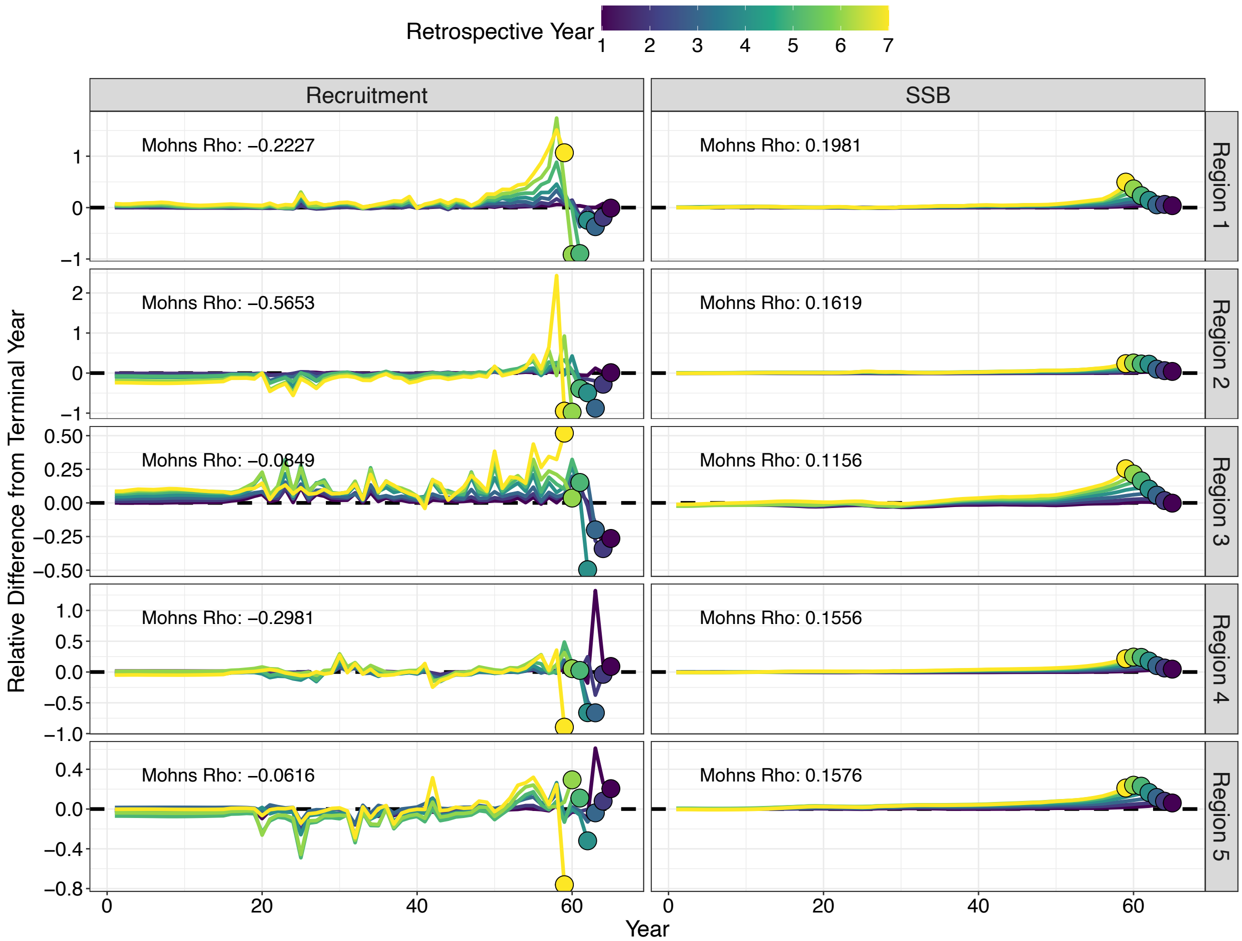

Figure 3D.47. Retrospective analysis from the five-region model. Column panels represent time series of recruitment and spawning stock biomass (*SSB*) as data are sequentially removed. Row panels denote the modeled regions, where region 1 represents the Bering Sea, region 2 represents the Aleutian Islands, region 3 represents the western Gulf of Alaska, region 4 represents the central Gulf of Alaska, and region 5 represents the eastern Gulf of Alaska. Mohn's rho values are shown in the upper left of each panel.

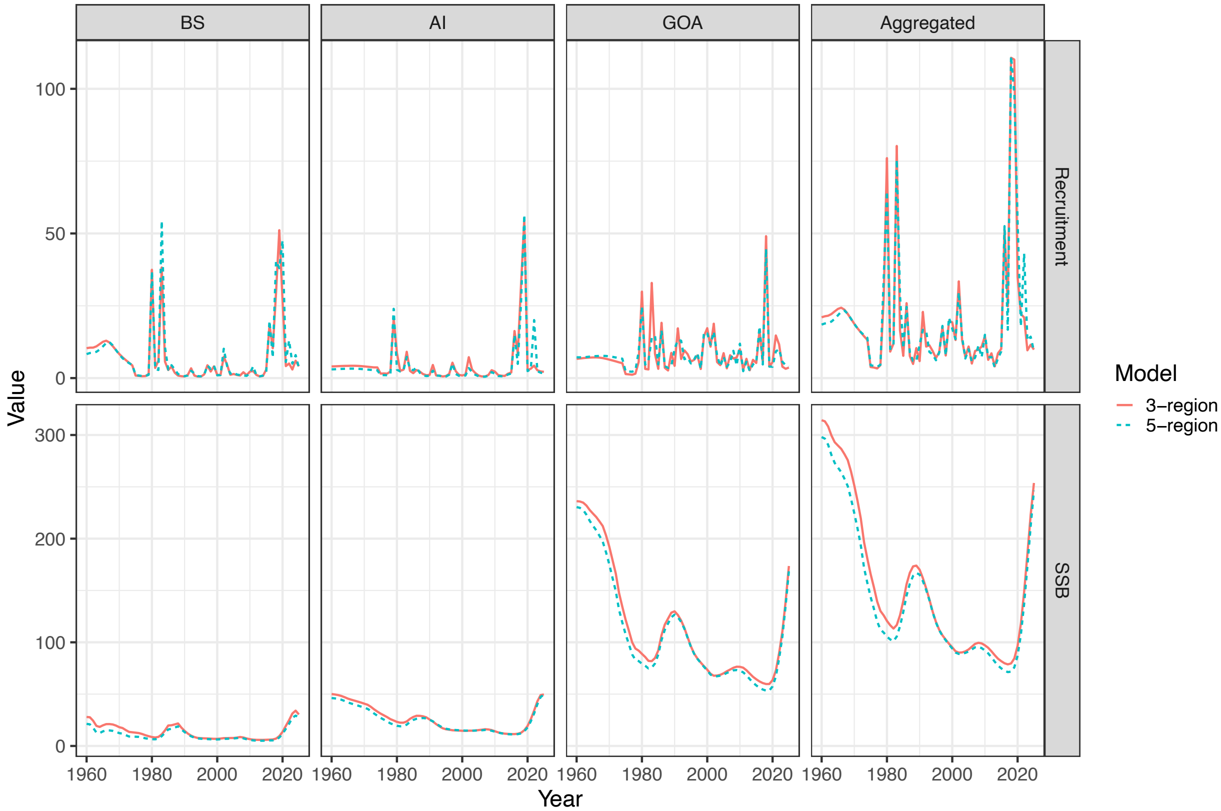

Figure 3D.48. Comparison of time-series of recruitment and spawning stock biomass (*SSB*) between the three-region and five-region spatial models. Time-series estimates for the Gulf of Alaska from the five-region model are summed up across the western, central, and eastern Gulf of Alaska for comparison with the three-region model.

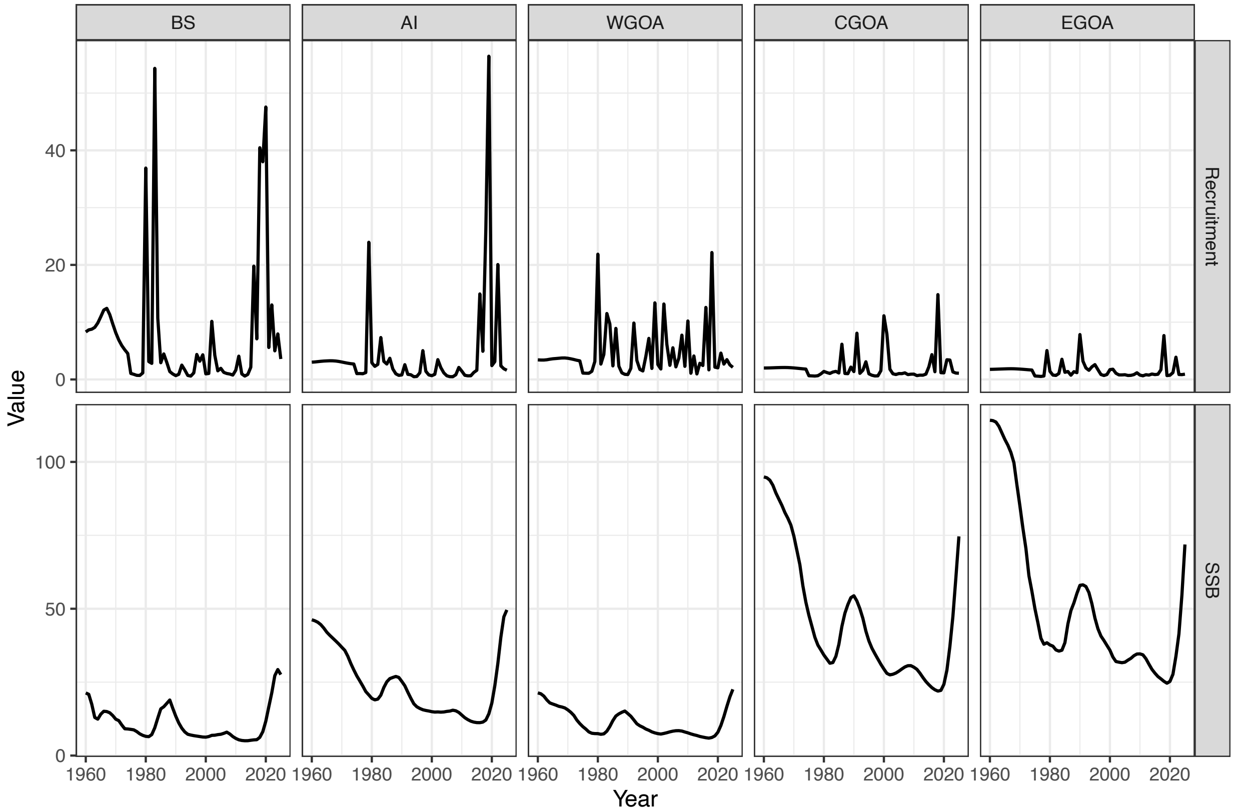

Figure 3D.49. Time-series of recruitment and spawning stock biomass (*SSB*) for the five-region spatial model.